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A Dissertation Phase-I Review -1 Report

On

GIS AND REMOTE SENSING-BASED SMART VILLAGE INFRASTRUCTURE PLANNING AND ROAD NETWORK OPTIMIZATION FOR RURAL TELANGANA

Submitted in partial fulfillment of the requirements for the degree

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Master of Technology
IT- (ARTIFICIAL INTELLIGENCE & ROBOTICS)

by

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CERTIFICATE

This is to certify that the Dissertation Phase-I titled "**GIS and Remote Sensing-Based Smart Village Infrastructure Planning and Road Network Optimization for Rural Telangana**" is the bonafide work carried out by **PENDYALA ARCHANA (160125763008)**, a student of M.Tech (AI & R) Semester-III at Chaitanya Bharathi Institute of Technology (A), Hyderabad, affiliated to Osmania University, Hyderabad, Telangana (India), during the academic year 2026-2027, submitted in partial fulfillment of the requirements for the degree in Master of Technology (AI & R) from **Department of Information Technology** and that this work has not formed the basis for the award previously of any other degree, diploma, fellowship or any other similar title.

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ABSTRACT

Rural road and public-service planning in Telangana needs methods that connect individual houses and priority facilities to the existing road network at low model cost, while exposing how errors in remote-sensing outputs change the resulting plan. This Phase-I report documents the problem, a completed literature review of 35 IEEE Transactions papers (2020–2026), each with its metadata verified against the DOI registry and its methodology extracted, plus five foundational references, the datasets used, and a proposed GIS and remote-sensing pipeline for the village of Damarakunta (Siddipet district). The pipeline detects building candidates from high-resolution imagery, assembles a 10 m cost surface from land cover and slope, and plans routes with a two-stage generate-and-select procedure in which facilities are routed first, houses are routed onto roads and previously planned routes, and shared edges are counted once.

Three contributions are framed. C1 studies how segmentation and candidate errors propagate into road plans. C2 is a two-stage planner with shared-edge credit. C3, transfer to other villages, is stated only as planned work. A*/least-cost paths, multi-criteria cost surfaces and facility prioritisation are used as established techniques and are not claimed as novel.

Preliminary, model-derived results are reported for Damarakunta: 278 building candidates, 165 inside the village boundary and 163 eligible as house targets, plus five facilities (two schools, a health sub-centre and two Gram Panchayat headquarters points; the anganwadi is excluded). With a 25° slope limit, all 163 houses and all five facilities were routed; sharing edges reduced the unique house-route length from 6.75 km (independent routes) to 5.07 km in the two-stage network-aware plan (about 24.9%, i.e. 1.68 km) and to 5.23 km in the roads-only shared-edge reference (about 22.6%). These numbers are planning-level, have no field validation, and do not establish legal access, ownership, passability or construction feasibility. Limitations and the Phase-II plan are set out explicitly.

The planned experiments were then run on the real Damarakunta data. Removing the building penalty changes the unique house network from 5,069 m to 4,705 m (-7.2%) and the routes that cross other building footprints from 28 to 56. Across 20 synthetic omit/add perturbation runs (10% and 20% of targets, five seeds each) every placed target was still routed and the unique length per routed target stayed between 30.6 m and 33.9 m (baseline 31.1 m). The perturbations are sensitivity probes, not measured segmentation error.

Keywords: remote sensing; semantic segmentation; building extraction; GIS; least-cost path; rural road planning; shared network; uncertainty propagation; Telangana.

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1. INTRODUCTION

1.1 Background and Motivation

Telangana's villages are served by a mix of state, panchayat and unclassified roads, and the household-level link between a house and the nearest usable road is rarely documented in a form that planners can analyse. High-resolution satellite imagery now makes it possible to produce building candidates and land-cover maps for a whole village, and GIS tools make it possible to turn those layers into candidate access routes. Deep-learning segmentation of remote-sensing imagery has advanced quickly for both buildings and roads [1], yet most published work stops at pixel- or object-level accuracy. How a missed or spurious building changes which routes are proposed, how much new network is needed and whether the result is connected is far less studied.

1.2 Problem Statement

Given (i) a set of building candidates produced by a segmentation pipeline, (ii) an existing road layer, (iii) a set of priority public facilities and (iv) a land-cover and slope-based cost surface, the problem is to propose a connected set of access routes from each target to the existing network that (a) respects hard terrain and land-cover constraints, (b) counts shared route segments once, and (c) reports how sensitive the plan is to errors in the target set. The deliverable is a planning-level proposal, not a construction design.

1.3 Research Gap

The reviewed literature (Chapter 2) concentrates on extracting buildings and roads. It rarely (i) checks vector-network connectivity of the result, (ii) compares planners on the same targets and cost surface, (iii) reports unique network length rather than summed route lengths, or (iv) propagates segmentation uncertainty into planning decisions. Benchmark results from urban or global datasets are also not evidence of rural performance [2, 3].

1.4 Aim and Objectives

Aim: to develop and evaluate, at planning level, a GIS and remote-sensing pipeline that proposes shared rural access networks for houses and public facilities in Damarakunta and quantifies sensitivity to segmentation errors.

Objectives

O1. Review 35 IEEE Transactions papers (2020 to 2026) on remote-sensing extraction, GIS and route planning, and identify the research gaps.

O2. Assemble and document the datasets and their provenance for the study village, including the five priority facilities (a primary school, a high school, a sub-centre and two Gram Panchayat headquarters points; the anganwadi is excluded), with source, count and selection reason for each.

O3. Produce building candidates with quality classes and a 10 m cost surface, using only road segments clipped to the village boundary and without interpolating missing slope or land-cover values.

O4. Implement a two-stage planner with shared-edge accounting and connectivity checks.

O5. Evaluate route length, unique length, constraint exposure, building crossings and connectivity, with automated validation against independent layers.

O6. Study the effect of target-set perturbations and building-penalty variants (C1).

O7. Define the Phase-II plan, including whether transfer to other villages (C3) is supportable.

1.5 Scope of the Project

The study area is the village of Damarakunta, Siddipet district, Telangana (projected CRS EPSG:32644, WGS 84 / UTM zone 44N). The scope is rural, planning-level and model-derived. It excludes legal access and ownership determination, passability and traffic assessment, structural or geometric road design, monetary costing, and field survey. Model-cost values are ordinal and are not currency.

1.6 Expected Contributions

Table 1 separates established techniques from the proposed contributions and states the current evidence status.

Table 1. Established techniques, proposed contributions and status.

Item	Description	Status
Established	A*/Dijkstra least-cost paths [4, 5], multi-criteria GIS analysis [6], direction-dependent least-cost routing [7], facility prioritisation. Not claimed as novel.	Used as is
C1	Propagation of segmentation/candidate errors (omission, addition, penalty choice) into route plans.	Preliminary; synthetic perturbations only
C2	Two-stage generate-and-select planner: facilities first, houses onto roads ∪ facility routes, shared-edge credit.	Implemented; preliminary results

Item	Description	Status
C3	Transfer of the pipeline to other villages.	Planned; not supported yet

2. LITERATURE SURVEY AND CRITICAL REVIEW

This chapter reviews 35 IEEE Transactions papers published between 2020 and 2026, and five foundational references. The review is complete for this scope, and the 35 papers meet the guide's requirement of at least 30 Transactions papers. For every paper the title, all authors, journal, year, volume, pages or article number and DOI were verified against the DOI-registry record (Crossref) on 29 October 2026, each DOI link leads to the IEEE Xplore page, and the objective, method, key findings and limitations were extracted and recorded; Appendix A lists the verified details of the 35 papers. Tables 2 and 3 give the full comparison, sorted by year with the newest first. Findings and limitations in Table 3 are a project-oriented critical assessment of each paper's published methodology and not necessarily statements by the authors. Numerical scores are quoted exactly as stated in the paper abstracts (nine of the 35 abstracts state scores) and, for five more papers whose open-access preprints could be read, as stated in the preprint text; these preprint figures may differ from the published IEEE version. The other entries say that no score is stated in the abstract, or that the abstract could not be retrieved (three papers), and no figure is quoted from memory. The datasets used by each paper were not recorded in the survey file, so a dataset column is omitted rather than guessed. The searches did not return an IEEE Transactions paper on rural village road planning from remote-sensing building targets, which is the gap this project addresses, but this is a statement about the searches made and not a proof that none exists.

2.1 Composition of the Review

All 35 works are IEEE Transactions papers: 30 in IEEE Transactions on Geoscience and Remote Sensing, 2 in IEEE Transactions on Image Processing, 2 in IEEE Transactions on Intelligent Transportation Systems and 1 in IEEE Transactions on Power Systems, all published between 2020 and 2026. By theme, they cover road extraction and topology, building extraction, segmentation with sparse labels and uncertainty, foundation-model and domain-adaptation methods, multi-objective route planning and GIS-based network planning. The planning themes are thinly covered in IEEE Transactions, which is why established least-cost-path references are kept as foundational references.

Table 2. Reviewed literature (identification).

S.No	Authors, year	Title	Venue	Class	Ref.
1	Y. Xing et al., 2026	CLIP2RS: Leveraging Pretrained Vision-Language Model for Semantic Segmentation of Remote Sensing Images	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[8]
2	J. Fu et al., 2026	HyperR3SNet: Leveraging Hyperbolic Space and Vision Foundation Models for Remote Sensing Semantic Segmentation	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[9]
3	X. Wang et al., 2025	Local–Global Sparse Transformer for Road Extraction From Remote Sensing Imagery	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[10]
4	P. Liu et al., 2025	CGCNet: Road Extraction From Remote Sensing Image With Compact Global Context-Aware	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[11]
5	S. Wang et al., 2025	Semi-Supervised Semantic Segmentation for Remote Sensing Images via Multiscale Uncertainty Consistency and Cross-Teacher–Student Attention	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[12]
6	Q. Cao et al., 2025	Open-Vocabulary High-Resolution Remote Sensing Image Semantic Segmentation	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[13]
7	L. Zhu et al., 2025	A Multi-Objective Path Planning Framework in Off-Road Environments Based on Deep Reinforcement Learning	IEEE Transactions on Intelligent Transportation Systems	IEEE Trans.	[14]
8	Y. Wang et al., 2024	A Multiscale and Multidirection Feature Fusion Network for Road Detection From Satellite Imagery	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[15]
9	Q. Li et al., 2024	A Review of Building Extraction From Remote Sensing Imagery: Geometrical Structures and Semantic Attributes	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[1]
10	W. Wang et al., 2024	Hierarchical Shared Architecture Search for Real-Time Semantic Segmentation of Remote Sensing Images	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[16]
11	H. Guo et al., 2024	Building-Road Collaborative Extraction From Remote Sensing Images via Cross-Task and Cross-Scale Interaction	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[17]
12	R. Yang et al., 2024	Occlusion-Aware Road Extraction Network for High-Resolution Remote Sensing Imagery	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[18]
13	Y. Wang et al., 2024	Extracting Building Footprints in SAR Images via Distilling Boundary Information From Optical Images	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[19]
14	Q. Tang et al., 2024	Enhancing Building Footprint Extraction With Partial Occlusion by Exploring Building Integrity	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[20]
15	D. Zheng et al., 2024	Weakly Supervised Building Extraction From High-Resolution Remote Sensing Images Based on Building-Aware Clustering and Activation Refinement Network	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[21]
16	Z. Wang et al., 2024	A Creative Weak Supervised Semantic Segmentation for Remote Sensing Images	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[22]
17	J. Li et al., 2024	RoadCorrector: A Structure-Aware Road Extraction Method for Road Connectivity and Topology Correction	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[23]
18	Z. Zheng et al., 2024	Pareto Improver: Learning Improvement Heuristics for Multi-Objective Route Planning	IEEE Transactions on Intelligent Transportation Systems	IEEE Trans.	[24]
19	Z. Wang et al., 2024	A Practical Urban Distribution Network Planning Method With Geographic Information System	IEEE Transactions on Power Systems	IEEE Trans.	[25]
20	R. Iizuka et al., 2024	Frequency-Based Optimal Style Mix for Domain Generalization in Semantic Segmentation of Remote Sensing Images	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[26]
21	C. Liang et al., 2024	Single Domain Generalization Method for Remote Sensing Image Segmentation via Category Consistency on Domain Randomization	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[27]

S.No	Authors, year	Title	Venue	Class	Ref.
22	A. Ma et al., 2023	Domain Adaptive Land-Cover Classification via Local Consistency and Global Diversity	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[2]
23	J. Hu et al., 2023	LGNet: Location-Guided Network for Road Extraction From Satellite Images	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[28]
24	X. Zhu et al., 2023	Semantics and Contour Based Interactive Learning Network for Building Footprint Extraction	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[29]
25	J. Zhang et al., 2023	Road Topology Extraction From Satellite Imagery by Joint Learning of Nodes and Their Connectivity	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[30]
26	Q. Xu et al., 2023	Road Extraction With Satellite Images and Partial Road Maps	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[31]
27	L. Ding et al., 2022	Adversarial Shape Learning for Building Extraction in VHR Remote Sensing Images	IEEE Transactions on Image Processing	IEEE Trans.	[32]
28	Q. Li et al., 2022	Building Footprint Generation Through Convolutional Neural Networks With Attraction Field Representation	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[33]
29	Y. Sun et al., 2022	CG-Net: Conditional GIS-Aware Network for Individual Building Segmentation in VHR SAR Images	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[34]
30	Y. Zhou et al., 2022	BOMSC-Net: Boundary Optimization and Multi-Scale Context Awareness Based Building Extraction From High-Resolution Remote Sensing Imagery	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[35]
31	Y. Liu et al., 2022	Multiregion Scale-Aware Network for Building Extraction From High-Resolution Remote Sensing Images	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[36]
32	H. Jung et al., 2022	Boundary Enhancement Semantic Segmentation for Building Extraction From Remote Sensed Image	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[37]
33	B. Li et al., 2022	POI Detection of High-Rise Buildings Using Remote Sensing Images: A Semantic Segmentation Method Based on Multitask Attention Res-U-Net	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[38]
34	B. Zhang et al., 2022	Curriculum-Style Local-to-Global Adaptation for Cross-Domain Remote Sensing Image Segmentation	IEEE Transactions on Geoscience and Remote Sensing	IEEE Trans.	[39]
35	J. Mei et al., 2021	CoANet: Connectivity Attention Network for Road Extraction From Satellite Imagery	IEEE Transactions on Image Processing	IEEE Trans.	[40]

Table 3. Reviewed literature: objective, method/key findings (with project relevance) and limitations.

S.No	Objective / outcome	Method, key findings and relevance	Limitations and gaps
1	Adapts a pretrained vision-language model using local/global alignment and text prompts for remote-sensing segmentation.	Relevant to foundation-model adaptation and class semantics in satellite imagery. Quantitative score: not stated in the abstract.	Natural-image priors and prompt choices may introduce domain/class errors; it is not direct evidence for rural infrastructure planning.
2	Adapts self-supervised vision foundation models with multiscale cross-attention and hyperbolic pixel-level loss for low-annotation segmentation.	Recent direction relevant to the user's DINOv2/foundation-model segmentation interest and cross-domain transfer concerns. Reported result: mean IoU 67.60% (iSAID), 55.86% (LoveDA), 80.07% (Potsdam) and 96.46% (Vaihingen).	Very recent results need independent reproduction and target-domain evaluation; published benchmark scores cannot be assumed to transfer to Damarakunta.
3	Combines local sparse attention and compressed global sparse attention with coarse-to-fine feature enhancement.	Relevant to road structures that require both local detail and long-range continuity. Quantitative score: not stated in the abstract.	Results on benchmark datasets need not transfer to Damarakunta; graph topology must be evaluated separately.
4	Uses a compact global-context block to improve road continuity while limiting computational complexity.	Directly relevant to connected road extraction and resource-efficient inference. Quantitative score: abstract not retrievable from the DOI registries.	Stronger raster connectivity is not equivalent to a verified, traversable GIS network.

S.No	Objective / outcome	Method, key findings and relevance	Limitations and gaps
5	Uses multiscale uncertainty consistency and cross teacher-student attention to exploit unlabeled imagery.	Relevant to uncertainty-aware segmentation and reducing the cost of dense annotation. Reported result (open-access preprint version): mean IoU of 74.62% (ISPRS-Potsdam) and 50.97% (LoveDA) with 5% labels, against 72.01% and 44.20% without the proposed modules.	Model uncertainty must be calibrated and evaluated; pseudo-label consistency is not a substitute for reference ground truth.
6	Introduces an open-vocabulary segmentation framework designed for orientation and scale variation in high-resolution remote-sensing images.	Relevant to flexible class definitions and possible future handling of diverse land-cover classes. Quantitative score: abstract not retrievable from the DOI registries.	Open-vocabulary output reliability for small rural objects and project-specific classes requires direct validation.
7	Plans paths for intelligent vehicles in off-road terrain with several objectives, including terrain-related failure risk and weather.	Actor-critic deep reinforcement learning on a traversability map with meteorology-driven heuristics; reported to beat standard DRL and non-DRL planners in efficiency and convergence stability. Relevant to terrain-aware multi-objective path cost. Quantitative score: not stated in the abstract.	Plans one vehicle path; it does not plan a network over many targets, and it is not an infrastructure-planning method.
8	Uses directional strip convolutions, multiscale residual features and a combined decoder to reduce fragmented road predictions.	Relevant to road completeness and robustness across image resolutions. Quantitative score: not stated in the abstract.	Improved raster extraction must still be converted to and audited as a connected road graph.
9	Reviews building extraction across optical and SAR imagery, including footprints, building attributes, height and change-related tasks.	Useful for positioning the building-detection stage and understanding geometric and semantic challenges. Quantitative score: not stated in the abstract.	A review does not establish which model is best for the project's sensor or rural settlement; local benchmarking is required.
10	Uses neural architecture search with a lightweight backbone and multiscale fusion to balance segmentation accuracy and speed.	Relevant when processing large imagery under limited compute and memory. Reported result: 54.5% mIoU (LoveDA), 87.8% (Potsdam) and 84.1% (Vaihingen) at 132.7 FPS; 72.6% mIoU at 164.0 FPS (CityScapes) and 72.3% mIoU at 186.4 FPS (CamVid).	Speed/accuracy trade-offs and LoveDA results are not evidence of Damarakunta performance or downstream planning correctness.
11	Jointly extracts buildings and roads using cross-task and cross-scale feature interactions.	Highly relevant to the project's coupled building/road layers and the spatial relationship between settlement objects and roads. Reported result (open-access preprint version): F1 higher by 3.21% (buildings) and 3.38% (roads) than the baseline (81.85% and 71.03%).	Joint extraction still does not solve route optimization, target prioritization, legal access or graph connectivity verification.
12	Introduces regular and stochastic occlusion-aware decoder components for road regions obscured by dense occlusions.	Addresses a common source of road discontinuity in overhead imagery. Quantitative score: not stated in the abstract.	Occlusion-aware segmentation may still hallucinate or miss segments; independent topology and reference checks remain essential.
13	Uses optical-image boundary information in a teacher-student distillation design for SAR building extraction.	Shows a way to exploit complementary modalities to improve boundaries where SAR interpretation is difficult. Reported result: F1 higher by 4.3 to 7.2 points and IoU higher by 4.9 to 8.0 points than the strong baseline on MSAW.	Requires suitable paired or aligned modalities and cannot be assumed applicable to an optical-only workflow.
14	Uses image enhancement and a building-integrity branch to improve footprints under partial occlusion.	Relevant to buildings partly hidden by trees, shadows or other structures. Reported result: IoU +1.49%, +1.16%, +2.26% and F1 +0.82%, +0.70%, +1.22% over HD-Net on WHU, Inria and AIRS.	Benchmark gains do not establish local accuracy; ambiguous merged footprints still need review and error tracking.
15	Uses building-aware feature clustering and activation refinement to generate better pseudo-masks from weaker labels.	Relevant to reducing labeling effort when preparing rural reference data. Reported result: IoU of 0.8556, 0.8163 and 0.7797 on the three datasets.	Weak supervision can produce uncertain pseudo-labels; these cannot be treated as ground truth for C1 evaluation.
16	Combines text-guided feature activation and Segment Anything Model refinement for weakly supervised remote-sensing segmentation.	Relevant to reducing dense annotation needs and exploring foundation-model assistance. Quantitative score: not stated in the abstract.	The method is weakly supervised and task/data dependent; outputs require independent label-based evaluation.

S.No	Objective / outcome	Method, key findings and relevance	Limitations and gaps
17	Corrects broken road surfaces and inaccurate topology in road extraction from high-resolution imagery.	Three stages: multi-branch road segmentation, energy-based connectivity refinement and topology correction into a vector network; improvements reported on several public datasets. Relevant to vector road graphs and gaps in them. Reported result: F1 improved by 3.3% to 4.5% and IoU by 2.0% to 5.1% over state-of-the-art methods.	Public benchmark data; whether corrected connections exist physically is not checked, and no route planning is done.
18	Learns improvement heuristics that approximate the Pareto front for multi-objective route planning problems such as logistics routing.	Population-based deep reinforcement learning model (Pareto Improver); the abstract reports that it outperforms other state-of-the-art methods on several multi-objective routing problems. Shows that multi-objective routing is an active area. Quantitative score: not stated in the abstract.	Vehicle-routing type problems with no terrain cost surface, shared infrastructure or uncertain targets.
19	Plans an urban power distribution network by combining GIS data with a hybrid optimisation model.	Extracts the street graph from GIS data, computes shortest paths between nodes, then uses Clark and Wright savings plus mixed-integer programming; reports that shortest-path distances cut investment by more than 30% compared with Euclidean or Manhattan distances in an urban Chinese case. Supports using path distances rather than straight lines in network planning. Reported result: computation more than 1000 times faster than MILP, and an investment decrease of 30% or more with shortest-path distance instead of Euclidean or Manhattan distance.	Urban power network, not rural roads; inputs are street data, not uncertain remote-sensing targets.
20	Improves generalisation of land-cover segmentation to unseen regions, sensors and seasons.	Frequency-based optimal style mix (FOSMix) randomises image style in the frequency domain with a consistency regulariser. Relevant to the transfer question behind C3. Quantitative score: not stated in the abstract.	Reports segmentation accuracy under domain shift, not the effect on a downstream plan.
21	Trains one segmentation model from a single source domain so that it works on unseen target domains.	Category Consistency on Domain Randomization (CCDR) randomises texture and style and adds a consistency loss across source and generated domains. Relevant to the case where no labelled target village is available. Quantitative score: not stated in the abstract.	Segmentation benchmark results only; downstream plan quality is not studied.
22	Introduces local-consistency/global-diversity regularization for unsupervised domain adaptation of land-cover mapping.	Relevant to the domain-shift problem between benchmark imagery and an unseen rural target village. Quantitative score: not stated in the abstract.	Land-cover adaptation results do not prove building-level or road-planning transfer; target-domain validation remains necessary.
23	Adds a road-location prediction task and location-guided decoder to improve road continuity.	Directly relevant because road connectivity is a structural requirement, not just a pixel-classification score. Quantitative score: not stated in the abstract.	Extracted connectivity still requires vector graph checks, endpoint snapping and comparison against authoritative roads.
24	Combines semantic and contour information in an interactive-learning approach for building-footprint extraction.	Supports attention to object contours and the possibility of targeted human review of ambiguous footprints. Quantitative score: not stated in the abstract.	Interactive/manual correction increases annotation effort and needs a documented protocol for reproducibility.
25	Builds a road network as a graph by predicting road nodes and the connectivity between them instead of skeletonising a segmentation mask.	NodeConnect, trained jointly on nodes and a connectivity map; compared with segmentation methods on DeepGlobe and RoadTracer with pixel and topological precision and recall. Relevant to the topology of the road layer a planner needs. Quantitative score: not stated in the abstract.	Topology is scored on benchmark data; the method does not link houses or facilities to the network.

S.No	Objective / outcome	Method, key findings and relevance	Limitations and gaps
26	Extracts roads from satellite images together with partial, incomplete road maps (for example from OpenStreetMap) instead of from scratch.	Two-branch Partial-to-Complete network with a gated self-attention module and a missing-part loss; IoU of 70.71% on SpaceNet and 75.52% on OSM data are reported. Close to the project situation of an incomplete TGRAC road layer. Reported result: IoU of 70.71% (SpaceNet) and 75.52% (OSM).	Completes road maps; it does not decide how houses should be connected, and rural Indian roads are not evaluated.
27	Introduces adversarial shape learning and a shape regularizer to improve building footprint geometry and segmentation.	Highlights the importance of object-level geometry metrics in addition to pixel accuracy; relevant to house/building target quality. Reported result (open-access preprint version): mean IoU higher by 2.73% (Inria) and 3.26% (MAS) than the baseline, and by 1.51% over the second-best method on Inria.	Footprint quality still depends on imagery, labels and domain characteristics; it does not evaluate downstream road-planning consequences.
28	Uses attraction-field representations with CNNs to generate building footprints from remote-sensing images.	Provides a geometry-aware alternative to plain pixel segmentation, relevant to separating and delineating adjacent buildings. Quantitative score: abstract not retrievable from the DOI registries.	Accurate footprint extraction alone does not resolve ambiguous building identity, entrances, legal access or route connectivity.
29	Uses GIS-aware conditioning for individual-building segmentation in very-high-resolution SAR imagery.	Shows how prior spatial/GIS information can assist individual-object delineation; relevant to combining imagery and geographic context. Reported result (open-access preprint version): F1 of 75.08% for DeepLabv3-CG on Berlin TerraSAR-X data, F1 and IoU 4.40% and 5.45% higher with complete footprints than with sensor-visible segments, and a mean building-height error of 2.39 m.	SAR-specific method and assumptions may not transfer directly to optical imagery or rural Telangana without evaluation.
30	Combines boundary optimization and multiscale context for building extraction.	Addresses boundary quality and scale variation, both important when converting building predictions to vector footprints. Quantitative score: not stated in the abstract.	Segmentation improvements do not by themselves quantify how missed, merged or false buildings alter route plans.
31	Models multiple regions and scales to handle buildings with varying sizes and layouts.	Relevant to rural settlements containing small houses and larger public buildings. Quantitative score: not stated in the abstract.	Generalization to different sensors, resolutions and rural morphologies must be measured; no network-planning evaluation is implied.
32	Combines edge cues with segmentation through a boundary-enhancement module that can be used with encoder-decoder networks.	Supports improving footprint edges across multiple building datasets and architectures. Quantitative score: not stated in the abstract.	Sharper boundaries do not guarantee correct instance separation or a real-world accessible connection.
33	Combines multitask attention Res-U-Net building-shape extraction with offset vectors and auxiliary POI information.	Illustrates how building geometry and semantic/point-of-interest information can be combined. Reported result: building-roof F1 77.05% and IoU 63.55%; whole-building F1 79.02% and IoU 66.05%, the best among the baseline models.	Targets high-rise buildings and auxiliary delivery data; transfer to rural houses and public facilities requires caution.
34	Adapts segmentation models across domains for very-high-resolution remote-sensing images.	Curriculum-style local-to-global adaptation using an entropy-based patch score and semantic- and entropy-level domain classifiers. Relevant to cross-village transfer of segmentation. Reported result (open-access preprint version): best segmentation accuracy of 48.06% and 52.03% in the two adaptation directions.	Needs target-domain images and does not evaluate network planning.
35	Extracts roads from satellite imagery while keeping them connected, because pixel-based methods often fragment roads where buildings or trees occlude them.	Connectivity attention network with strip convolutions and a connectivity attention module; the abstract reports state-of-the-art results on SpaceNet and DeepGlobe. Relevant to producing a connected base road layer. Quantitative score: not stated in the abstract.	Benchmark imagery; connectivity of the output is not tested against a planning network and there is no rural Telangana evidence.

2.2 Critical Discussion

2.2.1 Road extraction and topology

Road-extraction work targets fragmented predictions, occlusion and weak connectivity [28, 15, 18, 10, 11, 40, 30, 23]. These papers improve the raster road layer. They do not establish that the vector graph is connected or that it joins houses to the road network, which is what a planner needs. Topology-oriented methods represent the road layer as a graph or correct its connections [30, 23], and one method uses incomplete existing road maps as an extra input [31]; this is close to the incomplete TGRAC layer, but the output is again a road map and not a set of house connections.

2.2.2 Building extraction and instance quality

Building studies stress boundary precision, multi-scale context, shape regularisation and occlusion handling [32, 33, 35, 36, 37, 29, 20], with surveys in [1]. They motivate the quality classes (plausible, review, unresolved merge) used here: ambiguous or merged blobs should not automatically become confirmed houses. Detection of a footprint is also not evidence of an entrance, ownership or legal access. SAR-based building work [34, 19] is less relevant to optical rural imagery but shows that GIS priors can support building segmentation.

2.2.3 Building–road interaction

Joint extraction [17] indicates that the two layers share context. Extracting both layers is still distinct from deciding which targets are connected, generating candidate routes, ranking them under constraints and verifying connectivity.

2.2.4 Domain adaptation, benchmarks and foundation models

Land-cover models are sensitive to sensor, geography, season and label distribution [2]. Benchmarks such as LoveDA [3] cover urban and rural scenes, but their scores are not evidence of performance on Damarakunta. Recent vision-language, open-vocabulary and foundation-model adaptations [22, 13, 8, 9] are promising for scarce labels but need target-domain evaluation. Cross-domain adaptation and domain-generalisation methods for remote-sensing segmentation [26, 39, 27] aim at unseen regions, sensors and seasons, which is the situation C3 would meet in a new village; they report segmentation accuracy and not the effect of transfer on a downstream network plan.

2.2.5 Sparse labels and uncertainty

Sparse, weakly supervised and semi-supervised methods [21, 22, 12] reduce annotation cost, and uncertainty estimation [12] points towards confidence-aware decisions. In this project,

perturbation experiments are synthetic sensitivity probes; they do not measure real segmentation-error propagation, which needs reference labels.

2.2.6 Planning and optimisation

Least-cost path planning [4, 5, 7] and multi-criteria GIS analysis [6] are mature, and are used here as established tools. The reviewed perception papers rarely evaluate downstream planning decisions: how target errors change routes, what each route adds to the unique network, and whether comparisons use equivalent targets and costs. Learning-based multi-objective route planning [24], terrain-aware multi-objective off-road path planning [14] and GIS-based network planning with shortest-path distances [25] show that route cost, terrain and network design are studied in IEEE Transactions, but they address vehicle routes or urban power networks. Among the papers reviewed, none evaluates a village road network with shared-edge credit under uncertain building targets.

2.3 Alignment of the Reviewed Literature with the Pipeline

This section connects the 35 reviewed IEEE Transactions papers to the architectural choices of the pipeline. The word used is motivate, not prove: the papers report results on other datasets and tasks, and none of them evaluates a rural road-planning pipeline. Table 4 pairs each implemented component with the papers that motivate it, states what those papers support, and states what the project does not claim.

Table 4. Pipeline components, the reviewed papers that motivate them, and what is and is not claimed.

Pipeline component (as implemented)	Motivating papers	What the papers support	What is not claimed
Segmentation of building footprints: DINOv2-Base encoder (768 channels) with FPN fusion (256), an ASPP module and a residual decoder (class StrongDINOv2Seg); the detection stage is reused as a verified input	[13, 8, 9]; multi-scale context [35, 36, 15, 16]	Pretrained vision foundation encoders and multi-scale context modules are an active and reported-effective direction; for example HyperR3SNet reports mean IoU of 67.60%, 55.86%, 80.07% and 96.46% on iSAID, LoveDA, Potsdam and Vaihingen [9], and a searched real-time network reports 54.5% mIoU on LoveDA [16].	The model is not retrained or re-scored in this report; no reference labels exist for Damarakunta, so its accuracy there is not measured, and its weights provenance is undocumented.
Footprint quality classes (plausible, review, unresolved merge) instead of one confirmed building class	Boundary and shape: [32, 35, 37, 20]; review [1]	Boundary precision, shape regularisation and occlusion are recognised weaknesses of segmentation outputs, so ambiguous or merged blobs should not automatically become houses.	No shape regulariser is implemented; the classes are rule-based and were checked only against independent layers.
Local layers and benchmark: LoveDA for the model background; TGRAC boundary, roads, parcels and facilities and Google Open Buildings v3 for local checks (Chapter 4)	Domain shift: [2, 26, 39, 27]; LoveDA [3]	Segmentation accuracy drops across sensors, regions and seasons, so benchmark scores do not transfer to a new village and local verification is needed.	Agreement with Open Buildings or TGRAC is consistency, not accuracy.
Road layer: TGRAC road lines clipped to the village boundary; connectivity checked on the planning grid and on the lines	[40, 30, 23, 31]; [28, 18]	Road connectivity and topology matter for downstream use, and incomplete existing road maps are a recognised input problem; the TGRAC layer is incomplete near the boundary.	No road extraction is performed; the papers improve raster road maps and do not join houses to a network.

Pipeline component (as implemented)	Motivating papers	What the papers support	What is not claimed
Cost surface and least-cost routing: ordinal cost on a 10 m grid, 8-connected Dijkstra search with a binary heap (A* is not used; NetworkX is used only in one connectivity diagnostic; STRtree spatial indexes serve footprint tests)	Foundations [4, 5, 7, 6]; [24, 14, 25]	Cost-weighted path planning is established; planning with a realistic path distance instead of Euclidean or Manhattan distance reduced investment by 30% or more in a distribution-network study [25]; multi-objective route planning is studied in [24, 14].	Dijkstra and A* with an admissible heuristic return the same least-cost path; the choice affects speed only, which is not a constraint on a grid of 293,328 cells.
Two-stage generate-and-select planning: K = 3 candidate routes per target (Stage A), then selection by marginal cost with shared-edge credit and leaf re-routing (Stage B)	[24, 14, 25]	Generating candidates and then choosing among them under an objective trade-off is consistent with improvement-heuristic and multi-objective planners; sharing infrastructure lowers network cost.	Novelty is not claimed for shortest-path search or for sharing edges; the claim is the fair, unique-length accounting on one target set.
Target-set perturbation (C1) and building-penalty variants	[21, 22, 12]	Sparse-label and semi-supervised work shows that predictions are uncertain and may be partial, which motivates testing how target changes propagate.	The omit and add runs are synthetic probes, not measured segmentation error.

The chain from literature to results is as follows. The reviewed segmentation papers motivate the detection stage and its quality classes (Section 2.2.2); Chapter 4 documents the datasets that stand in for ground truth (LoveDA for the model background, and the TGRAC and Open Buildings layers for local checks); and the planner then turns the 163 eligible targets and five facilities into a shared network. On these inputs the two-stage plan has a unique house-route length of 5,069 m against 6,753 m for independent routes (a saving of 24.9%), and Chapter 9 reports the sensitivity of that result. The papers themselves are not evidence for these numbers.

3. RESEARCH GAP AND PROPOSED CONTRIBUTIONS

From Chapter 2, six gaps are carried into the project.

1. **Perception-to-planning propagation.** Evaluate how building and road errors change target service status, routes, unique network length and constraint exposure, separating synthetic sensitivity from label-based evaluation (C1).
2. **Network validity.** Beyond raster IoU/F1, check snapping, connected components, attachment chains and unserved targets.
3. **Fair comparison.** Compare planners on the same targets, cost surface and slope limit; report served/unserved targets and the unique-edge union (C2).
4. **Rural-domain evidence.** Do not infer rural performance from urban benchmarks.
5. **Uncertainty-aware decisions.** Test whether uncertain predictions should trigger review or lower confidence in the plan.
6. **Practical outputs.** Report length and model cost separately from money, and flag parcels, compounds, agriculture and water as potential impacts needing verification.

These gaps are not proof of novelty. The project does not claim novelty for shortest-path search, least-cost paths, A* or multi-criteria analysis, which are established [4, 5, 7, 6]. The contribution claimed is narrower: C1 measures how changes in the building-target set propagate into route selection, network length and exposure, and C2 generates alternatives first and selects among them with unique-length and shared-edge accounting. C3 (five-village transfer) is planned and has not started. Whether this combination is new must still be argued against prior art beyond the 35 papers reviewed, which remains open for Phase-II.

Table 5. Research-gap matrix: what the reviewed literature does, what is missing and how the project responds.

Gap	What the reviewed papers do	What is missing	Project response
G1 Perception to planning (C1)	Building and road extraction methods are scored by pixel or topology metrics [33, 35, 36, 40, 30].	How target errors change routes, unique length and exposure is not evaluated.	C1 omit, add and merge runs on the real village (synthetic perturbations).
G2 Network validity	Topology-aware road methods repair connections in the road layer [30, 23, 31].	Connection of houses to roads, components and attachment chains is not checked.	Connectivity audit; Table 17.
G3 Fair comparison (C2)	Route planners compare objectives or cost on single or vehicle routes [24, 14]; GIS path distances help network design [25].	No comparison of independent routes against a shared network on the same targets and cost surface.	Independent, roads-only and length-only baselines; shared-edge credit; C2 ablation.
G4 Rural-domain evidence (C3)	Domain adaptation and generalisation address unseen regions [26, 39, 27]; LoveDA mixes urban and rural scenes [3].	Effect of transfer on a downstream village plan is unknown.	C3 is planned and has not started.
G5 Uncertainty-aware decisions	Uncertainty estimation and sparse-label learning [12, 21].	No test of whether uncertain targets should trigger review of the plan.	Quality classes and penalty variants P1 to P3; label-based test is future work.

Gap	What the reviewed papers do	What is missing	Project response
G6 Practical outputs	Reviewed papers report accuracy, not planning proxies.	Parcel, compound and land-cover exposure beside length and ordinal cost.	Impact flags (exposure counts, not ownership).

4. DATASET SOURCES, DESCRIPTION AND JUSTIFICATION

Record counts below come from the project files and audit documents inspected for this report; where a count or source is not documented, it is stated as such and not estimated. Table 6 summarises the datasets. Table 7 gives the justification and role of each.

Table 6. Datasets: source, coverage and records.

Dataset	Source / link	Coverage, resolution, CRS	Records and attributes	Limitations
High-resolution image mosaic (`damarakunta_esri_highres_mosaic_0p6m.tif`)	Esri tiles exported by the project's own script; Esri World Imagery (`https://www.arcgis.com`)	Damarakunta; 0.6 m; georeferencing not independently verified	One mosaic raster. RGB imagery	Capture date and licence not documented in the project; same imagery family seen by the model, so visual checks are not independent
Building candidates (`buildings_v8_full.gpkg`, `candidate_footprints_for_review_v4_1.gpkg`)	Project pipeline output (`extract_buildings_v8.py`; segmentation model)	Damarakunta; polygons; EPSG:32644	278 candidates; 165 inside the village boundary; 163 eligible targets after excluding two below 10 m^2. Quality class: 71 plausible, 32 review, 62 unresolved merge (165 inside)	Model-derived; model-weights provenance unresolved; no field or official confirmation
Existing roads (`tgrac_road_network.gpkg`)	Telangana TGRAC REST query (Telangana GIS portal; link/licence not recorded in the project)	Village area; vector lines; EPSG:32644	43 line features. Road type, sub-type, name, owner, length	Generalised and incomplete (a visible paved road in the east is missing); licence/terms not recorded
Priority facilities (`damarakunta_priority_facilities.gpkg`)	Built from TGRAC school and health layers by `prepare_damarakunta_priority_facilities.py`, plus two Gram Panchayat headquarters points from TGRAC layer 10 (see Section 4.3); the anganwadi record is excluded	Village; points; EPSG:32644	5 facilities in the final layer (a primary school, a high school, a sub-centre, two Gram Panchayat headquarters points), all priority group 1. Facility id, name, type, group	Identity rests on project-supplied compound layer and TGRAC attributes; not confirmed against an official registry
Cadastral parcels (`damarakunta_cadastral.gpkg`)	TGRAC cadastral query	Village; polygons; EPSG:32644	794 parcels (earlier project check). Parcel geometry/attributes	Some parcels were invalid and repaired; parcel intersection is not ownership or access
Land cover (`damarakunta_worldcover_10m.tif`)	ESA WorldCover (`https://esa-worldcover.org`); the project notes record 2021 v200, not stored in the raster	10 m raster	One raster. Tree, shrub, grass, crop, built-up, bare, water	Version not stored in the raster; 10 m classes can miss small rural objects
Terrain (`damarakunta_slope_10m.tif`)	Derived from a DEM of about 30 m; DEM product not documented in the files read	10 m slope grid	One raster. Slope (degrees)	Slope is resampled from 30 m data and has no true 10 m terrain detail; 961 gaps inside the boundary are not interpolated
Cost surface and blocked mask	Derived by `create_damarakunta_cost_surface.py`	10 m rasters	Two rasters. Ordinal cost; blocked cells	Model-cost values are not currency
Open Buildings v3 (comparison only)	Google Open Buildings (`https://sites.research.google/gr/open-buildings/`)	Local cache	Large local CSV cache. ML-derived polygons	ML product, not ground truth; licence use not recorded in the project
LoveDA (benchmark reference only)	[3]; `https://github.com/Junjue-Wang/LoveDA`	Urban and rural scenes, 0.3 m	As reported in the dataset paper (not re-counted here). Seven land-cover classes	Not the target village; used only to frame transfer limits

Table 7. Why each dataset was selected and how it is used.

Dataset	Reason for selection	Use
Image mosaic	Sub-metre imagery is needed to see individual rural roofs and compounds; it is the only imagery available for the whole village.	Actual village planning
Building candidates	Provide the house targets; quality classes make segmentation uncertainty explicit and support C1.	Actual village planning; sensitivity experiments
TGRAC roads	Official-source reference layer for the existing network that targets must connect to.	Actual village planning
Priority facilities and administrative utilities	Define the first-stage targets for the shared network (C2); the Gram Panchayat headquarters points add administrative service destinations.	Actual village planning
Cadastral parcels	Allow post-hoc parcel-crossing proxies; not used as constraints.	Reference and impact proxy
Land cover	Provides per-class traversal cost and blocks water.	Cost surface
Slope	Provides terrain penalty and hard slope limits (25°, 10° scenarios).	Cost surface
Open Buildings v3	Independent ML product for overlap context.	Reference comparison
LoveDA	Public benchmark with urban and rural domains; frames domain-shift limits for C3.	Benchmark reference only

4.1 Verified Counts and Units

Table 8 separates image counts, vector features, candidate buildings, routing targets, raster cells and route alternatives, and gives the source of each count. Counts marked as read from a file were measured on 29 October 2026; LoveDA figures come from its Zenodo record and arXiv abstract. Where the licence or acquisition date is not recorded, the table says so instead of assuming one.

Table 8. Dataset inventory with units, verified counts, source and status.

Update of 31 October 2026: at the project owner's request the anganwadi (facility 1702300, Damarakunt-I) is excluded from the final planning layers, tables and maps because its TGRAC coordinate lies 371 m from the nearest detected footprint (inputs/EXCLUSIONS.md). Two Gram Panchayat headquarters points are added as administrative utilities (Section 4.3). All facility counts and results in this report refer to the final five-facility layer; the verified four-facility baseline run is used only for the reproduction check.

Dataset	Unit	Count and size	Source and link	Status
LoveDA benchmark	Images; annotated objects	5,987 images at 0.3 m from Nanjing, Changzhou and Wuhan (Zenodo record); 166,768 annotated objects (arXiv abstract)	Wang et al., NeurIPS 2021 Datasets and Benchmarks; doi 10.5281/zenodo.5706578; arXiv 2110.08733; github.com/Junjue-Wang/LoveDA; CC BY 4.0 per the Zenodo record	Public documentation checked 29 Oct 2026. Segmentation development only; not Damarakunta evidence.
Damarakunta imagery	Raster pixels	11,296 x 7,171 pixels, 3 bands, 8-bit, 0.6 m, EPSG:32644 (read from the file); bounding box 29.16 km2	Esri World Imagery, 11 MapServer export requests on 4 Oct 2026	Capture date unknown; licence unresolved; enquiry not sent.
Village boundary	Vector feature	1 polygon, 365 vertices, 18.466 km2, EPSG:32644 (read from the file)	TGRAC Administrative Information System, MapServer layer 27, filter Village = DAMARAKUNTA, Mandal = MARKOOK, District = SIDDIPET (tgrac.telangana.gov.in)	Equals the stored query response (IoU 1.0, 0 m2 difference); no independent boundary; licence not stated.
TGRAC road network	Vector features	43 line features, 93,496.76 m; 41,789.51 m (44.70 percent) lies outside the polygon and 20 of 43 features reach outside; clipped with gpd.clip() to 50 line parts, 51,707.25 m, all inside, which are the roads routed in the final scenarios	TGRAC, as recorded in the project (Schools MapServer layer 20)	Completeness and connectivity unverified; licence not stated.
Building candidates	Polygon instances	278 candidates (135 plausible, 59 review, 84 unresolved merge); 165 footprints inside the boundary; 163 routing targets (101 + 62)	Project extraction, buildings_v8_full.gpkg (model-derived)	Automatic and unverified; no reference buildings; not a census.
Facility targets	Point features	5 in the final layer (2 schools, 1 sub-centre, 2 Gram Panchayat HQ points); the TGRAC anganwadi is excluded	TGRAC school and health layers; TGRAC layer 10 (Gram Panchayat headquarters)	Identity and coordinates unverified; sub-centre has two records 235.7 m apart.
Cadastral parcels	Polygon features	794 parcels (EPSG:4326; 7 invalid geometries repaired in the working copy); union area 18.2051 km2; overlap with the boundary 18.0616 km2; boundary uncovered 0.4045 km2; parcels outside 0.1435 km2 (0.79 percent)	TGRAC cadastral layer (source URL not recorded in the files inspected)	Geometric comparison only (97.81 percent boundary coverage); not ownership or legal access.
School compounds	Polygon features	2	Creation method unknown	Not independently confirmed.
Land cover	Raster cells	679 x 432 cells at 10 m; 178 invalid cells inside the polygon	ESA WorldCover (esa-worldcover.org); project notes record 2021 v200, CC BY 4.0	Version not stored in the raster; classification accuracy for this village not assessed.
DEM and slope	Raster cells	DEM 227 x 144 cells at 30 m, 564.3 to 625.8 m; slope resampled to the 10 m grid; 961 slope gaps inside the polygon, not interpolated (limitation of the 30 m DEM)	File n17_e078_1arc_v3 (SRTM-type name); product and licence not recorded	Not true 10 m terrain; source unverified.
Planning grid and cost surface	Raster cells	432 x 679 = 293,328 cells; 184,646 inside and 108,682 outside the polygon; cost 2.09 to 12.0; 99.31 percent valid inside	Derived (create_damarakunta_cost_surface.py)	Ordinal cost; provisional weights.
Route alternatives and routes	Line features	319 Stage A alternatives for 163 targets; 168 selected routes (163 houses, 5 facilities)	Derived (Stage A and B)	Planning proxies, not roads.

4.2 Role of Each Source and What LoveDA Does Not Show

The sources play complementary roles. Imagery gives building candidates; the TGRAC boundary limits the study area; the TGRAC roads give the existing network that routes must reach; land cover and slope give the cost surface; facilities give the first planning group; cadastral parcels and school compounds are used only to flag possible exposure, not ownership. Each also has a weakness: the imagery has no recorded date, the roads are

incomplete and extend beyond the boundary, the building candidates are unverified, the DEM is coarse, and the parcel layer does not show legal access.

LoveDA is a public land-cover benchmark with 5,987 images at 0.3 m from three Chinese cities, with urban and rural domains [3]. It is useful for developing and comparing segmentation methods and for framing the domain-shift question behind C3. It is not a Telangana dataset, and no LoveDA score is evidence of how well buildings or roads are found in Damarakunta. Model results from LoveDA development are therefore not reported here as Damarakunta accuracy.

4.3 Administrative and Governance Utilities

Besides schools and the health sub-centre, a village baseline needs the administrative and governance points that residents use: the Gram Panchayat office, the MeeSeva citizen-service centre and the Directorate of Electronic Service Delivery (ESD) service and Aadhaar centre. They are treated as primary administrative utility points of the village baseline, and the same layer structure is meant to be reused for other villages. Table 9 gives the source, count, position, checks and status of each. Only the Gram Panchayat headquarters points could be sourced.

Table 9. Administrative and governance utilities: source, count, position and verification status.

Utility	Source and count	Position (lat, lon)	Checks against other layers	Status
Damarakunta Gram Panchayat headquarters	TGRAC Administrative Information System, MapServer layer 10 (Gram Panchayat Head Quarter, point layer); 1 point	17.706404 N, 78.753015 E	Inside the village boundary; table latitude/longitude equals the geometry; 0.7 m from a detected footprint (an unresolved-merge blob, building not identified); 3.4 m from an Open Buildings polygon	Included in the facility layer
Kashireddypally Gram Panchayat headquarters (second GP whose revenue village is Damarakunta)	Same layer; 1 point	17.699092 N, 78.779906 E	Inside the boundary, in the eastern settlement; 0.0 m from a detected footprint; 2.5 m from an Open Buildings polygon	Included in the facility layer
MeeSeva centre	No source found: none of the 162 TGRAC map services (2,019 layers) has a MeeSeva, ESD, Aadhaar or citizen-service layer, and OpenStreetMap has no such feature in the village area; 0 points	not located	none possible	Not included; coordinate and source needed
Directorate of ESD (Aadhaar/service) centre	Same search; 0 points	not located	none possible	Not included; coordinate and source needed

Selection reason. The Gram Panchayat is the village's governance body and a destination for residents; its headquarters location is published by TGRAC in the same service as the village boundary, which makes the source official, queryable by village and reusable for every other

village in the template. Verification means agreement with independent layers: both points lie inside the boundary and on or within 1 m of a detected building footprint and within 4 m of an Open Buildings polygon. This shows consistency, not that the building is the Panchayat office; TGRAC calls the point a headquarters location, and a visual or field check is still needed. The MeeSeva and ESD centres are not placed on a guess: in many villages the MeeSeva centre sits in the Panchayat office, but that is an assumption and is not applied. Coordinates for them can be added through inputs/admin_utilities_intake.csv once a source (the MeeSeva or UIDAI locator, a register or a field GPS reading) is available.

Facility counts used throughout this report: five facilities in the final layer (primary school, high school, sub-centre and two Gram Panchayat headquarters points). The TGRAC anganwadi (Damarakunt-I) is excluded because its coordinate lies 371 m from the nearest detected footprint (inputs/EXCLUSIONS.md); it is kept only as an excluded record. Note on the two Gram Panchayat points: the headquarters points of two Gram Panchayat jurisdictions, Damarakunta and Kashireddypally, both lie inside the Damarakunta village boundary (the second in the eastern settlement), so both administrative panchayat jurisdiction points are held in the spatial database and both are planning targets.

5. PROPOSED SYSTEM ARCHITECTURE

5.1 Overview

Figure 1 shows the architecture in nine stages and Figure 2 the flow from inputs through scripts to outputs. Section 5.2 describes each stage. Table 10 lists the 20 implementation modules, with each module's purpose, input, algorithm, output, evaluation and status. Status uses four labels: implemented, partial, planned and unverified. Nothing marked partial or planned is reported as complete.

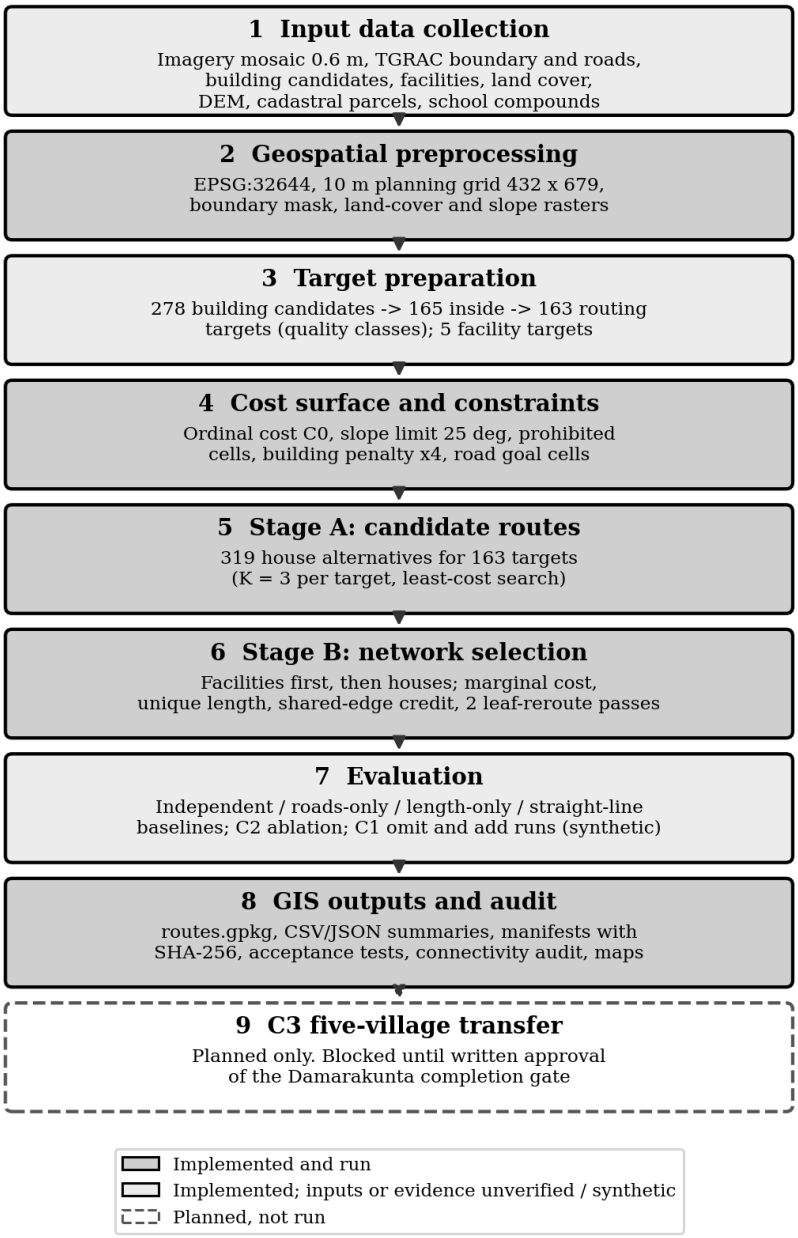


Figure 1. High-level architecture in nine stages. Dark gray: implemented and run; light gray: implemented but inputs or evidence unverified or synthetic; white with dashed border: planned.

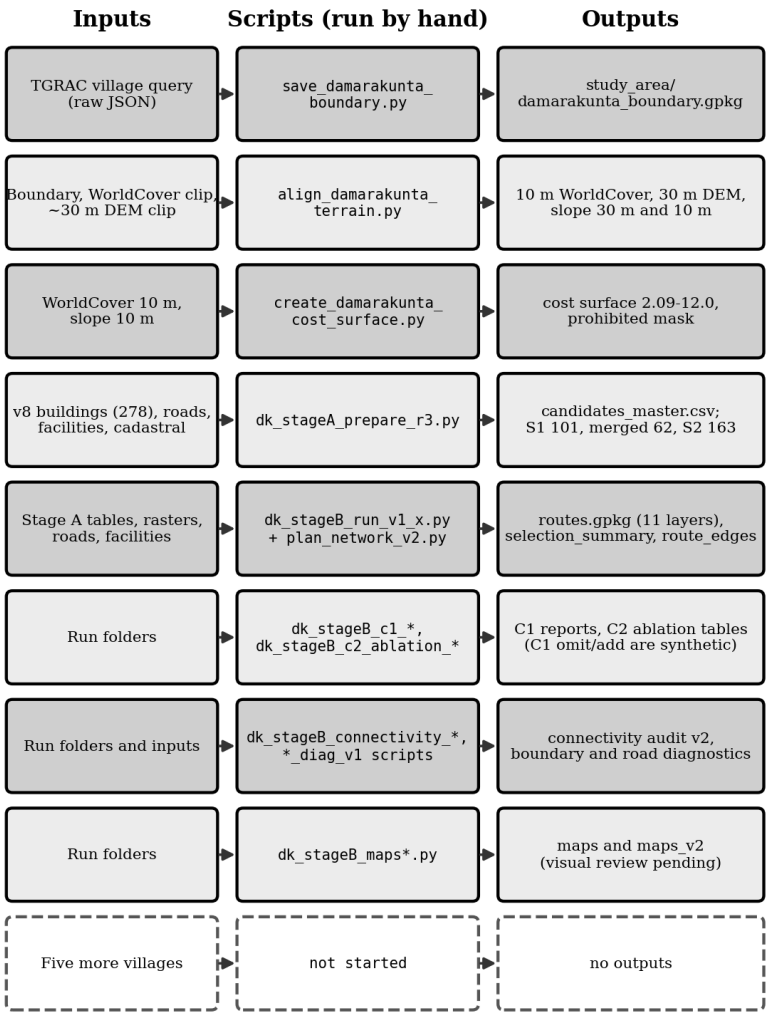


Figure 2. Detailed workflow: inputs, scripts and outputs. Shading as in Figure 1; every box corresponds to a script or output in the project folder.

Table 10. Module description (status as of this report).

#	Module	Purpose (input)	Algorithm	Output; evaluation	Status
M1	Imagery and GIS inputs	Assemble village data (input: Mosaic, TGRAC layers, parcels)	File inventory with hashes	Provenance table (34 items); evaluation: File existence, hashes	**Implemented**
M2	Source verification	State what is and is not verified (input: Provenance, scripts)	Evidence-level audit	Audit report; facility and footprint verification tables; evaluation: Evidence levels; no field or official confirmation reached	**Partial**
M3	Georeferencing	Common CRS and alignment (input: Mosaic, layers)	Reprojection to EPSG:32644	Aligned layers; evaluation: Visual overlay; georeferencing not independently verified	**Partial**
M4	Preprocessing, clipping	Village extent (input: Layers, boundary)	Clip, geometry repair	Clipped layers; evaluation: Validity checks; repaired cadastral geometry is provisional	**Implemented**
M5	Building detection	Candidate footprints (input: Mosaic)	Segmentation model, polygonisation, quality classes	278 candidates; 165 inside; evaluation: Internal GIS checks only; no reference labels; weights provenance unresolved	**Implemented** / **Unverified**

#	Module	Purpose (input)	Algorithm	Output; evaluation	Status
M6	Land cover and cost surface	Per-class cost (input: Land cover)	Class cost lookup	Cost raster, blocked mask; evaluation: Range checks	**Implemented**
M7	Existing roads	Network goals (input: TGRAC roads)	Rasterise to goal cells	43 road features; road goal cells in 12 components; evaluation: Counts, component sizes	**Implemented**
M8	House targets	Define targets (input: Footprints)	Area filter (≥10 m^2), scenario subsets S1/S2	163 targets (S1 101, S2 163); evaluation: Stage A: 51 PASS, 9 INFO, 0 WARN, 0 FAIL	**Implemented**
M9	Facility prioritisation	Stage-1 targets (input: TGRAC school/health/anganwadi)	Equal-weight priority group	4 facility targets; evaluation: Identity unverified; weights provisional	**Partial**
M10	Raster/vector conversion	Link grid and vectors (input: Cells, polylines)	Cell-centre rasterisation, polyline reconstruction	Route geometries; evaluation: CRS and validity checks	**Implemented**
M11	Slope and terrain	Terrain constraint (input: DEM slope)	Sobel slope from ~30 m DEM, bilinear to 10 m	Slope raster; evaluation: Limited by 30 m source	**Implemented** (limited)
M12	Multi-criteria cost surface	Combine land cover and slope (input: M6, M11)	Eq. (1)	Cost surface; evaluation: Baseline raster fixed; scenario C1 provisional	**Implemented**
M13	Candidate routes	Generate routes (input: Cost, targets, goals)	8-connected Dijkstra grid planner	Route per target; evaluation: Routed counts, reasons for failure	**Implemented**
M14	Feasibility and ranking	Select among options (input: Routes)	Constraint check, shortest-cost selection	Selected routes; evaluation: Hard constraints only; no ranking beyond cost	**Partial**
M15	Shared-edge credit	Count shared edges once (input: Routes)	Equal-share edge accounting, Eq. (3)	Unique length, credits; evaluation: Reproduction of single-stage reference	**Implemented**
M16	Graph and connectivity	Check network (input: Cell graph)	Disjoint-set union	Components, cycles; evaluation: Cycles 0; one house route not on main road component (under investigation)	**Implemented**
M17	House/facility connectivity	Check attachment (input: Routes, status)	Attachment classes road/facility/house	Attachment counts, chain depth; evaluation: Max chain depth 8	**Implemented**
M18	Metrics	Summaries (input: All outputs)	Totals, unique lengths, strata, impact proxies	CSV tables; evaluation: Cross-run comparison	**Implemented**
M19	QC, sensitivity, uncertainty	Robustness (input: Runs)	Omit/add perturbations, penalty variants, validator	Validation report, tables; evaluation: Sandbox-tested; real-data runs **Planned**	**Partial**
M20	Maps and reports	Communicate (input: Outputs)	Vector maps, tables	Figures, report; evaluation: Visual check by the author required	**Partial**

5.2 Stage-by-Stage Description

The nine stages below follow Figure 1. Each gives inputs, processing, outputs, validation, limitations and the execution status. Table 11 summarises them. Stages 1 to 8 were run on Damarakunta; Stage 9 has not started.

Table 11. Summary of the nine stages (inputs, processing, outputs, validation and status).

Stage	Inputs	Processing	Outputs	Validation	Status
1	Imagery, TGRAC layers, building candidates, facilities, cadastral, land cover, DEM	Store; re-project	Layers and hashes	Stage A input validation	Implemented; provenance unresolved
2	Boundary, land cover, DEM	10 m grid 432 x 679; slope; masks	Aligned rasters	Mask diagnostic reproduced counts	Implemented
3	278 candidates, 5 facilities	Quality classes; S1 101 + merged 62 = 163 targets	scenario_targets.csv	Stage A: 51 PASS, 9 INFO, 0 WARN, 0 FAIL	Implemented; targets unverified
4	Land cover, slope, prohibited mask, roads	Ordinal cost C0; 25 degree limit; building penalty x4; goal cells	Cost surface; 6,494 goal cells	Diagnostics reproduced counts	Implemented
5	163 targets, cost surface	K = 3 least-cost alternatives per target	319 alternatives	Geometry and identifier audit	Implemented

Stage	Inputs	Processing	Outputs	Validation	Status
6	Alternatives, facilities	Marginal-cost selection; shared-edge credit; 2 reroute passes	routes.gpkg (11 layers)	Acceptance tests; connectivity audit	Implemented
7	Run folders	Baselines; C2 ablation; C1 omit/add runs	Tables and reports	C2 R01 to R03 PASS (recorded)	Implemented; C1 synthetic
8	Run folders	Manifests; validators; maps	Manifests, audits, maps	Validator 396 PASS, 28 INFO; audit v2 2 FAIL	Implemented; map review pending
9	Five villages	Not started	None	None	Planned

5.2.1 Stage 1: Input data collection

Status. Implemented; provenance partly unresolved

Inputs. Esri World Imagery mosaic at 0.6 m (11,296 x 7,171 pixels, three bands, 8-bit, EPSG:32644, assembled on 4 October 2026 from 11 map-export requests); TGRAC village boundary (one polygon); TGRAC road lines (43 features); building candidates from the project's own extraction (278); five facility points (TGRAC school and health points and two Gram Panchayat headquarters points from TGRAC layer 10; the anganwadi record is excluded); cadastral parcels (794) and two school-compound polygons; a land-cover raster and a DEM clip.

Processing. Files are stored as GeoPackage or GeoTIFF and re-projected to EPSG:32644 where needed. No field data are collected.

Outputs. Layers under study_area/, reference/ and aligned/, with SHA-256 hashes recorded in each run manifest.

Validation. A read-only Stage A input validation checked CRS, grid alignment, geometry validity and counts.

Limitations. The imagery capture date is unknown and its licence is unresolved (an enquiry is drafted but has not been sent). The TGRAC licence, currentness and completeness are not verified. The DEM file name (n17_e078_1arc_v3) suggests an SRTM-type 1 arc-second product but the source is not recorded. The land-cover raster carries no version tag; WorldCover 2021 v200 is the project's recorded source.

5.2.2 Stage 2: Geospatial preprocessing and alignment

Status. Implemented and run

Inputs. Boundary polygon, land-cover clip and about 30 m DEM clip.

Processing. All layers are placed on one 10 m grid of 432 x 679 cells (293,328 cells) in EPSG:32644. Slope is computed from the 30 m DEM and resampled to 10 m. Cells whose centres lie outside the polygon are masked: 108,682 cells outside and 184,646 inside.

Outputs. Land cover at 10 m, DEM at 30 m, slope at 30 m and 10 m, prohibited mask, cost surface.

Validation. The boundary-mask diagnostic rebuilt the counts and the input hashes matched the Stage B manifest.

Limitations. Inside the polygon 178 cells have no land-cover value and 961 have no slope value. The cost-surface validation attributes 1,276 invalid inside cells to 178 land-cover gaps, 224 water cells and 874 slope gaps; the 961 and 874 slope counts come from different checks and are not reconciled here. Resampling a 30 m DEM does not give 10 m terrain accuracy. These gaps are a spatial limitation of the 30 m source data and were not filled by interpolation; the planner treats them as invalid cells. The boundary polygon has not been compared with an independent boundary.

5.2.3 Stage 3: Building-target and facility preparation

Status. Implemented; targets unverified

Inputs. 278 building candidates (135 plausible, 59 review, 84 unresolved merge; 10 further blobs rejected as mostly on no-data), the village boundary and five facility points (three school and health points, two Gram Panchayat headquarters points; the anganwadi is excluded).

Processing. Candidates inside the boundary (165 footprints) receive quality classes. Stage A builds two target sets: a default set of 101 and a merged-sensitivity set of 62, together 163 eligible house targets (scenario S2). Two of the 165 footprints (candidates 189 and 202) were not routed. Facilities form the first planning group.

Outputs. scenario_targets.csv (163 houses) and facility targets (5).

Validation. Stage A checks: 51 PASS, 9 INFO, 0 WARN, 0 FAIL; default plus merged targets equal the routed targets.

Limitations. Candidates are automatic detections with status AUTO_DETECTED_UNVERIFIED. No reference buildings or manual labels exist, so detection accuracy cannot be computed. The 163 targets are not a census of village houses. Facility identity is unresolved: the school layers are attribute-only, the Gram Panchayat

points are headquarters locations (the building is not identified), the sub-centre has two TGRAC records 235.7 m apart, and the anganwadi was excluded because no footprint lies within 371 m of its coordinate.

5.2.4 Stage 4: Cost surface and spatial constraints

Status. Implemented and run

Inputs. 10 m land cover and slope, prohibited mask, road lines.

Processing. Scenario C0 gives each valid cell an ordinal cost from land cover (range 2.09 to 12.0). Cells steeper than the slope limit (25 degrees in the main runs) and invalid or prohibited cells are impassable. Moves are 8-connected with corner cutting forbidden, and a step costs the mean of the two cell costs times the step length. Entering a penalised building cell multiplies the step weight by 4 (Section 6.5.2). Road lines are rasterised into goal cells: 8,061 cells are covered by road lines, 6,494 are passable goals and 1,567 are dropped as impassable.

Outputs. Cost surface, prohibited mask (37.49 percent of the grid, mostly outside the polygon) and goal cells.

Validation. Boundary and chain-cell diagnostics reproduced the counts above; the prohibited mask and the cost surface follow the same validity rule by construction.

Limitations. Costs are ordinal planning units, not rupees. The land-cover cost weights are provisional. The prohibited mask is a computational exclusion and not a legal restriction. Of the 12 road components in the grid, the main one has 6,149 cells (94.7 percent of goal cells); 1,564 road cells with invalid cost lie mostly outside the polygon (Section 9.5).

5.2.5 Stage 5: Stage A candidate-route generation

Status. Implemented and run

Inputs. 163 house targets, cost surface, goal cells.

Processing. For each target the planner finds up to K = 3 least-cost routes to the road goal cells. The first is the plain least-cost route; later ones are found after raising the cost of cells within 20 m of earlier alternatives (multiplier 3, then 8 if needed), and an alternative that shares more than 80 percent of its cells with an earlier one is rejected. Targets are snapped to a valid cell within 30 m and the search is limited to 1,500 m.

Outputs. Layer stageA_alternatives_houses with 319 alternatives for the 163 targets (so not every target has three), each labelled UNVERIFIED_PLANNING_ALTERNATIVE_NOT_A_ROAD.

Validation. The audit of the K3_P2 run found no null, empty or invalid geometry in the alternatives layer and matching target identifiers.

Limitations. Alternatives come from one cost surface and are therefore correlated. They are planning proxies, not designed roads.

5.2.6 Stage 6: Stage B network selection and shared-edge credit

Status. Implemented and run

Inputs. Stage A alternatives, facilities, road goal cells.

Processing. Facilities are routed first and their routes become attachable. Houses are then taken in fixed order and each is attached to the road or to an already selected route (attachment tolerance 15 m) by comparing candidate marginal cost. Two leaf-reroute passes re-evaluate end routes. Shared grid edges are counted once in the unique length and credited in the cost (Equation 3). Baselines are independent least-cost routes, a roads-only reference, a length-only control and straight lines.

Outputs. routes.gpkg with 11 layers, selection_summary.csv, route_edges.csv, shared_edge_accounting.csv and connectivity_checks.csv.

Validation. Per-run acceptance tests (primary: 48 PASS, 1 INFO; P3: 41 PASS, 2 INFO) and the connectivity audit (Section 9.5).

Limitations. Shared-edge credit counts identical grid edges only. The selection is greedy and is not claimed to be globally optimal. Facility weights are equal and provisional.

5.2.7 Stage 7: Evaluation, C1 uncertainty runs and C2 ablations

Status. Implemented; C1 runs are synthetic

Inputs. Run folders from Stage B.

Processing. Metrics: unique length (m), total length (m), shared length (m), unique and total ordinal cost, saving against the independent baseline (percent), footprint crossings and attachment class. C2 ablation varies alternatives per target (K = 1 or 3) and leaf-reroute passes (0 or 2). C1 omits or adds 10 and 20 percent of targets with five seeds each, and omits merged blobs, and measures footprint exposure.

Outputs. Tables in _final_main, _c1_report_v4 and _c2_abl_v1.

Validation. C2 ablation validation R01 to R03 PASS (read from c2_ablation_validation.csv); the 14 planner tests and the 49 C1 report checks are recorded in project notes and were not re-run here.

Limitations. The omit and add runs are seeded synthetic probes and not measured segmentation error. K1 and K3 give equal summary lengths at each pass setting, but identical geometry is not shown for every route. No claim of general superiority is made.

5.2.8 Stage 8: GIS outputs, validation and evidence packaging

Status. Implemented; map review pending

Inputs. Run folders and diagnostics.

Processing. Each run writes GeoPackage layers, CSV and JSON summaries, config_stageB.json, inputs_manifest_stageB.csv and outputs_manifest.csv with SHA-256 hashes. A read-only validator covered 30 runs (396 PASS, 28 INFO, 0 WARN, 0 FAIL). The connectivity audit v2 is a separate check and reports the C0002 attachment as a failure (Table 17).

Outputs. _final_main tables, _validation_main, maps and maps_v2, and the DELIVERABLES_DAMARAKUNTA_STAGE_B folder.

Validation. Manifest hash checks; P3 manifest 19 rows without missing files or hash mismatches (recorded in project notes).

Limitations. The four maps were checked for scale bar, north arrow, legend and CRS, but no figure is declared publication-ready and one C1 chart has a legend overlapping a label. The steps are run by hand and are not one automated pipeline.

5.2.9 Stage 9: C3 multi-village transfer

Status. Planned; not started

Inputs. Five additional villages (not selected).

Processing. Re-run Stages 1 to 8 on each village and compare planning outcomes with Damarakunta.

Outputs. None.

Validation. Not applicable.

Limitations. C3 is blocked until the project owner approves the Damarakunta completion gate in writing. No transfer result exists, and none is claimed.

6. DETAILED METHODOLOGY AND WORKFLOW

6.1 GIS Data Preparation and Georeferencing

All vector and raster layers are placed in EPSG:32644. Invalid cadastral polygons were repaired by a buffer-zero style repair; the repaired geometry is provisional and parcel-based proxies flag repaired parcels. The imagery georeferencing has not been independently verified, so positional offsets of a few metres cannot be excluded. A drift of about 0.9 m between map-grid north and true north is noted in the map annotation of the project figure.

6.2 Remote Sensing-Based Building and Land-Cover Processing

Building candidates are produced by a segmentation pipeline (`extract_buildings_v8.py`) and polygonised. Each polygon receives a quality class: *plausible*, *review* or *unresolved merge* (Figure 3). Candidates below 10 m^2 are excluded from targets. Land cover is a 10 m class raster (ESA WorldCover 2021 v200 according to the project notes; the version is not stored in the raster). These classes are model outputs and are not ground truth.

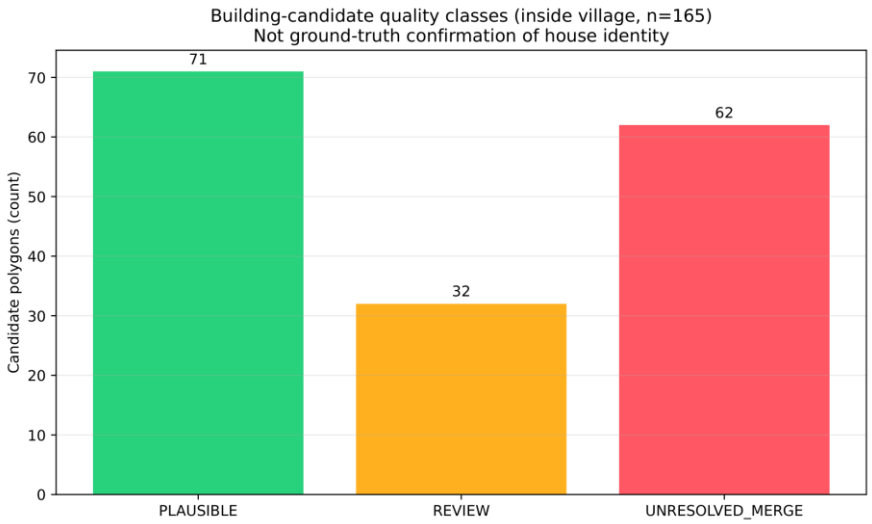


Figure 3. Building-candidate quality classes inside the village (n = 165). The classes are model-derived and do not confirm house identity.

6.3 Road Network Extraction and Existing Road Integration

The existing network is the TGRAC road layer (43 line features), which is generalised and incomplete. Road lines are rasterised to goal cells; the goal cells form 12 connected components (sizes 6149, 159, 62, 29, 28, 23, 16, 10, 9, 5, 2, 2). A target is "connected" only if its route ends on a goal cell. Road extraction from imagery is reviewed in Chapter 2 but is not performed in this phase.

6.4 Public-Service Facility and House Connectivity Planning

Five facilities (a primary school, a high school, one sub-centre and two Gram Panchayat headquarters points from TGRAC; the anganwadi is excluded) are Stage-1 targets with equal provisional weights. Houses in Stage 2 are routed onto the union of roads and facility routes; each route is classified as attached to a road, a facility route or another house route. The planner is an 8-connected grid Dijkstra search [5] (equivalent to A* with a zero heuristic [4]) run in a fixed order.

6.5 Multi-Criteria Route Generation and Route Selection

6.5.1 Cost surface

Let $c_{\mathrm{lc}}(x)$ be the land-cover cost and $s(x)$ the slope in degrees at cell x. The baseline cost is

$$c(x) = c_{\mathrm{lc}}(x) + 4\,\frac{\mathrm{clip}(s(x),\,0,\,10)}{10} \tag{1}$$

with c_{lc} = 8 (tree), 5 (shrub), 3 (grass), 6 (crop), 2 (built-up), 4 (bare). Water, no-data and invalid-slope cells and cells with s above the scenario limit (25° or 10°) are blocked. Costs are ordinal model units, not currency. The C1 scenario sets built-up cost to 3.0 (provisional).

6.5.2 Building penalty

Cells inside footprints of the penalised quality classes raise the edge weight by a factor m=4 when the planner enters them:

$$c'(x) = m\,c(x) \text{ for } x \in \bigcup_{b \in \mathcal{B}_q} \mathrm{footprint}(b);\quad c'(x) = c(x) \text{ otherwise} \tag{2}$$

where B_{q} is the set of buildings whose class is in q. The baseline uses all three classes; variants P1 (plausible only), P2 (plausible and review) and P3 (no penalty) are reported in Section 9.3. It is a cost, not a block: in the planner (plan_network_v2.py) the weight of a step into a penalised cell is multiplied by m.

6.5.3 Path cost and shared-edge credit

For a route k with edge set E_{k}, the planner minimises $\Sigma_{(\mathrm{i,j})\backslash \mathrm{in\ E_k}}\, d_{\mathrm{ij}}\ (c_{\mathrm{i}}+c_{\mathrm{j}})/(2)$, where d_{ij} is the step length (10 m or 10√2 m). This formulation was checked against the planner source (plan_network_v2.py), where the step weight is 0.5 ($c_{\mathrm{u}}+c_{\mathrm{v}}$) times the step length. With n_{e} the number of routes using edge e and ℓ_{e} its length,

$$L_{\text{unique}} = \sum_{e \in \bigcup_k E_k} \ell_e, \quad \text{credit}_k = \sum_{e \in E_k} \frac{\ell_e}{n_e}, \quad \sum_k \text{credit}_k = L_{\text{unique}} \tag{3}$$

The shared-edge saving is L_{ind}-L_{unique}, where L_{ind} is the unique length of independent single-target routes.

Algorithm 1. Two-stage generate-and-select planning (schematic)

Input: cost surface c, goal cells G (roads), facilities F, house targets H
1. Route each facility f ∈ F to G by Dijkstra; add its cells to G_F
2. for each house h ∈ H in fixed order:
3. route h to G ∪ G_F ∪ G_H; record attachment class
4. add the route's cells to G_H
5. Compute L_unique, credits (Eq. 3), components and attachment chains
6. Re-run with roads only as goals for the reference baseline

6.5.4 Planning-level versus real-world feasibility

A route being found means only that a cell path exists under the model constraints. It is not evidence of legal access, ownership, passability, construction approval or field feasibility.

6.6 Sensitivity and Uncertainty Design (C1)

Two perturbations are defined. Omit removes a random fraction (0.10 or 0.20) of scenario targets and their footprints, representing missed detections. Add inserts synthetic translated copies (15–50 m) of random buildings on valid cells at least 1 m from roads, representing spurious detections; synthetic additions are probes only. Seeds are fixed (20261029–20261033) so runs are reproducible. Both perturbations, with five seeds per setting, and the three penalty variants have been run on the real Damarakunta data; the results are in Sections 9.3 and 9.4. They measure sensitivity to synthetic target changes and not real segmentation error.

7. EXPERIMENTAL SETUP AND EVALUATION METRICS

Scenarios. S1 uses the 101 plausible and review candidates (71 plausible, 30 review among eligible targets); S2 adds 62 unresolved-merge candidates (163 targets). Slope limits of 25° and 10° are run for each, and one cost scenario with built-up cost 3.0 is run (C1).

Metrics. Routed and failed targets; total and unique route length; shared-edge saving; attachment counts and chain depth; connected components and cycles; slope and land-cover shares; penalised-cell length; number of other footprints crossed; length inside parcels flagged as repaired. Cost values are not comparable across cost scenarios.

Baselines. Independent single-target routes, a straight-line baseline, a length-only baseline and a roads-only reference that must reproduce the earlier single-stage result exactly.

Automated validation. A read-only validator checks required files, manifest hashes, target counts and identifiers, duplicates and missing targets, geometry validity, connectivity, failure reasons, regression against the reference and reproducibility of paired runs. Stage A passed 51 checks with 9 informational notes and 0 warnings or failures. The Stage B acceptance validator covered 30 runs (primary scenarios, penalty variants, perturbation runs and paired reruns): 396 PASS, 28 INFO, 0 WARN and 0 FAIL; it rates the one house route that ends on a disconnected road fragment as INFO. A separate connectivity audit (version 2) applies a stricter rule to the same route and reports it as a failure (Section 9.5). The two audits have different scopes and are not interchangeable.

Boundary clipping. In the final scenarios the TGRAC road lines were clipped to the village boundary with `gpd.clip()` before routing: of 93,496.76 m (43 lines), 41,789.51 m (44.70 percent; 20 lines reached outside) lay outside the boundary and was removed, leaving 51,707.25 m in 50 line parts that lie entirely inside the boundary; only these are used as road goals and for connectivity (`src/preprocessing/clip_roads.py`, `inputs/road_clip_report.csv`). The original road file is not edited. One local override is used for the health sub-centre only: the building penalty is lifted within 20 m of the sub-centre point, because the facility stands at the edge of its own footprint and the penalty forced its route to detour round that footprint across open ground; no other route is affected (runner extttdk_stageB_run_v1_6.py, option exttt–facility-exempt). **Environment.** Python 3.10, projected CRS EPSG:32644, 10 m grid. The 10 m slope is resampled from about 30 m DEM data and does not contain true 10 m terrain detail. Inside the boundary, 961 cells have no slope value and 178 cells have no land-cover value because of the coarse 30 m DEM and

land-cover source; these gaps are a spatial limitation of the source data. They were not filled by interpolation or any other synthetic estimate: the planner treats them as invalid cells that routes cannot cross.

8. IMPLEMENTATION AND CURRENT PROJECT PROGRESS

Table 12 records progress against the modules. The scripts were written in Python and executed on the project machine; the planner file was imported unmodified and its hash is recorded.

Table 12. Progress summary.

Component	Evidence	Status
Stage A target set	278 candidates, 165 inside, 163 eligible; S1 101, S2 163 (disjoint; S2 adds 62 unresolved merges); 51 PASS / 9 INFO / 0 WARN / 0 FAIL	**Implemented**
Stage B runs	S1 and S2 at 25° and 10°, plus C1 scenario; per-run acceptance checks have 0 FAIL; connectivity audit v2 reports one failed attachment (C0002)	**Implemented**
Reference reproduction	Roads-only reference reproduces the earlier single-stage run (unique lengths 5231.54 m and 6755.36 m; geometry difference 0.00 m)	**Implemented**
Penalty variants P1–P3	Run on real data for S2 at 25° (Table 15)	**Implemented**
Perturbation runs (omit/add)	20 runs on real data (final family): omit and add, 10% and 20%, five seeds each (Table 16); synthetic probes only	**Implemented**
Automated validator and final tables	33 runs audited: 432 PASS, 31 INFO, 0 WARN, 0 FAIL (acceptance validator, final family); final tables written	**Implemented**
Field validation, official confirmation	None	**Planned**
Literature review	35 IEEE Transactions papers (2020 to 2026): metadata verified against the DOI registry; methodology, findings and limitations extracted for every paper	Complete
Evidence-status items	Twelve open items tracked in Table 20	**Partial**

9. PRELIMINARY RESULTS AND DISCUSSION

All results are model-derived, preliminary and unvalidated in the field.

All figures in this chapter come from the final scenario family: 163 house targets and five facilities (two schools, one sub-centre and two Gram Panchayat headquarters points). The anganwadi is excluded (inputs/EXCLUSIONS.md); the MeeSeva and Directorate of ESD centres have no located coordinate and are not included (inputs/ADMIN_UTILITIES.md). Run run_20261102_141136.

9.1 Shared-network results

Table 13. Stage B results for S2 (25° slope limit, 163 house targets, 5 facilities).

Quantity	Value
Targets routed	163 / 163 houses; 5 / 5 facilities
Network-aware house routes, total length	5,069.04 m
Network-aware house routes, unique length	5,069.00 m
Independent baseline, total length	10,045.38 m
Independent baseline, unique length	6,753.10 m
Unique-length saving	1,684.10 m (24.9%)
Roads-only reference, unique length	5,229.28 m
Facility routes, unique length	249.31 m
Combined (houses + facilities) unique length	5,318.30 m
Attachment counts (houses): road / facility route / house route	95 / 6 / 62
Maximum chain depth; cycles	9; 0
Straight-line baseline length (2 infeasible)	8,092.30 m

Table 13 shows that sharing edges shortens the unique network, mainly for long routes: in the longest stratum (more than 20 m from a road, 91 targets) the mean route length is 45.3 m network-aware versus 99.8 m independent, and 59 of those 91 routes attach to another house route. In the two shorter strata (up to 10 m, 48 targets; 10 to 20 m, 24 targets) the means are nearly equal (9.6 vs 9.8 m and 20.2 vs 20.7 m).

Two saving percentages are quoted because they use different comparators against the same independent baseline (6,753.10 m unique): 22.6% is the roads-only shared-edge reference (5,229.28 m) and 24.9% is the two-stage network-aware plan (5,069.00 m). Table 12 covers five facilities: the primary school, the high school, the sub-centre and the Damarakunta and Kashireddypally Gram Panchayat headquarters points; both administrative panchayat jurisdiction points are held in the spatial database and routed, and the anganwadi is excluded. Roads were clipped to the village boundary with gpd.clip() before routing. Adding the two

Gram Panchayat headquarters points as facility targets shortens the house network by 160.3 m, because some houses attach to the facility routes; against the four-facility verified baseline (5,137.7 m of house routes) the change is 68.7 m.

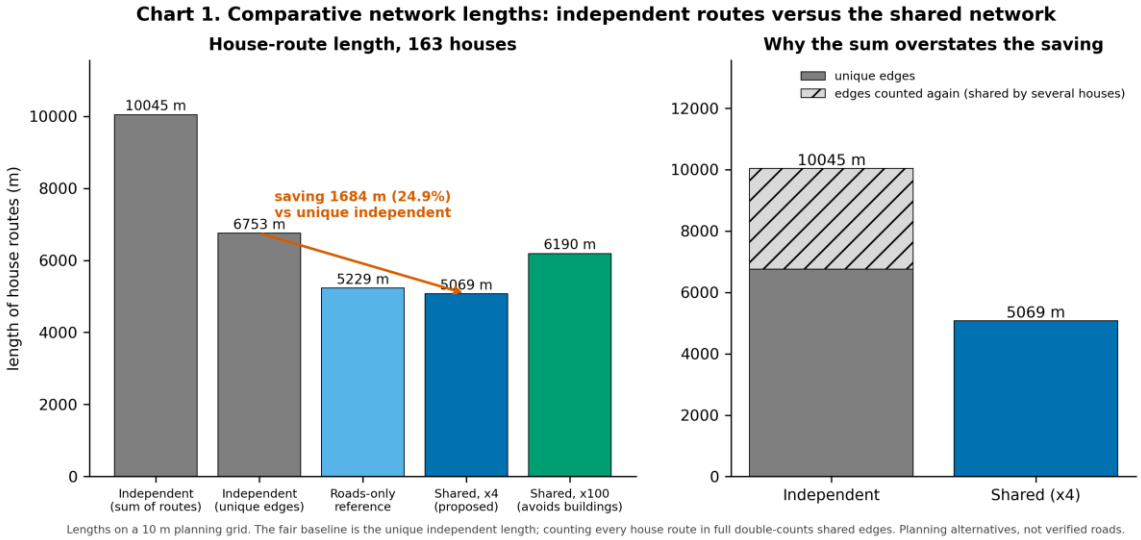


Figure 4. Unique grid-edge length: independent routes versus the shared network (chart 1 of the final package; model-derived, no field validation).

Table 14. Other runs (final scenario family).

Run	Routed (houses)	Network unique (m)	Independent unique (m)
S1, 25°	101 / 101	3,599.51	4,785.06
S2, 10°	161 / 163	4,858.85	6,447.84
S1, 10°	99 / 101	3,339.37	4,469.80
S2, C1 (built-up 3.0), 25°	163 / 163	5,029.47	6,940.42

At 10 degrees 1 house target could not be routed (TARGET_IN_HARD_CONSTRAINT_CELL_OR_OUTSIDE_GRID) because their start cells exceed the limit; this is a model constraint, not a statement about the buildings.

9.2 Facility routes and impact proxies

Facility routes are 16.9 m (MPPS primary school), 52.3 m (ZPHS high school; attached to 36181600902), 56.6 m (Damarakunta Gram Panchayat HQ), 56.1 m (Kashireddypally Gram Panchayat HQ) and 67.4 m (sub-centre). Planning facilities first reduces the house unique length by 160 m (S2) and 122 m (S1). Network-aware house routes: 28 routes cross another footprint (209 m) against 46 for independent routes, 60 for straight lines and 64 for length-only routes. Penalised-cell length is 601 m (network) versus 645 m (independent); route length on slopes above 10 degrees is 44 m and none exceeds 25 degrees; about 75 m lies

inside parcels that were invalid in the original cadastral layer and needs visual checking. These are proxies, not ownership or access findings.

9.3 Building-penalty variants

Table 15 compares the baseline penalty (all three quality classes, multiplier 4) with P1 (plausible only), P2 (plausible and review) and P3 (no penalty), all for S2 at the 25° limit; every run routes 163 of 163 houses. The shortest unique network is P3 (4,705.4 m, 7.2% shorter than the baseline 5,069.0 m); P3 has the most routes crossing other footprints (56, against 28 for the baseline), and P1 and P2 cross 47 and 45. The saving column compares each run with independent routes planned under the same penalty, so it is not comparable across rows as a length measure. Footprint classes are model-derived and a route that avoids them is not evidence that real compounds are avoided.

Table 15. Building-penalty variants (S2, 25° slope limit, 163 house targets).

Variant	Penalised classes	Unique length (m)	Saving (%)	Routes crossing other footprints	Length in other footprints (m)	Penalised-cell length (m)
Baseline	Plausible, review, unresolved merge (x4)	5,069.0	24.9	28	209.1	600.7
P1	Plausible (x4)	4,851.4	26.9	47	560.6	244.1
P2	Plausible, review (x4)	4,903.0	27.1	45	556.8	374.1
P3	None (x1)	4,705.4	32.0	56	660.1	0.0

9.4 Sensitivity to target-set perturbations

Table 16 summarises 20 perturbation runs on the S2 scenario. Every target that could be placed was routed, with 0 failures. The unique length per routed target stayed between 30.6 m and 33.9 m over the 20 runs, against 31.1 m for the unperturbed baseline (5,069.0 m over 163 targets). On targets common to the base and perturbed runs, the mean change in route length per run ranged from -2.5 m to +1.8 m and 95% to 99% of targets kept the same attachment class (road, facility or house route). The widest spread in total unique length is for 20% omission (3,991 m to 4,411 m), because removing a target can remove branches that other houses were sharing. In 9 of the ten add runs not all requested synthetic buildings could be placed, which the run checks log as a warning. These perturbations are synthetic sensitivity probes; they do not measure how real segmentation errors would change the plan.

Table 16. Perturbation runs, mean over five seeds (range in brackets).

Perturbation	Targets routed	Unique length (m)	Length per routed target (m)	Routes crossing other footprints	Same attachment class (%)
Add 10%	177.4	5,502.4 (5,393.2–5,548.8)	31.0	32.0	98
Add 20%	190.6	5,868.3 (5,822.3–5,916.1)	30.8	36.4	97
Omit 10%	147.0	4,684.4 (4,521.0–4,821.9)	31.9	24.2	97
Omit 20%	130.0	4,211.5 (3,991.4–4,410.9)	32.4	20.4	96

9.5 Connectivity

Two connectivity definitions are in use and they give different counts. The planner-level check (the route ends on the declared network and the cell graph is connected) holds for all 168 routed targets. The stricter check, that the route ends on the main road component of 6,149 cells, holds for 167 of 168 targets, or 162 of 163 houses. Both statements concern one target, C0002 (candidate 2, house H0001, class review): its route is 105.93 m long, starts 95.58 m from the road layer and ends on a 5-cell road component (component 10; component 11 in the 10 degree runs) that is cut off from the main component inside the planning grid. The target, route length and component are identical in the x4 and x1 runs, so the two penalty configurations do not differ in connectivity. Table 17 lists the evidence for each statement.

Table 17. Connectivity reconciliation: which run, target and definition each statement refers to.

No.	Run	Metric and definition	Observed	Evidence file	Reading
R1	S2_fac_C0_s25_final_x4 (primary, x4)	planner_connected_to_roads and cell_graph_connected, all 168 targets (163 houses, 5 facilities)	168 of 168 true	connectivity_checks.csv	Planner-level connection holds for every routed target.
R2	S2_fac_C0_s25_final_x4 (primary, x4)	reaches_main_road_component: chain terminal lies on the 6,149-cell main road component	167 of 168 (houses 162 of 163, facilities 5 of 5). The exception is target C0002 (candidate 2, house H0001, class review, 95.58 m from the nearest road); its route is 105.93 m and ends on road component 10 (5 cells)	connectivity_checks.csv; road_components.csv	Different definition from R1. One target ends on a fragment cut off from the main component inside the grid.
R3	S2_fac_C0_s25_final_x1 (x1)	Same two fields	168 of 168 and 167 of 168. Same target C0002, same component 10	connectivity_checks.csv; target_status.csv	The x1 and x4 runs agree on connectivity: 167 of 168 (all targets) and 162 of 163 (houses only) describe the same single target.
R4	S2_fac_C0_s10_final (10 degree limit)	Same fields	165 of 166 reach the main component; C0002 on component 11	connectivity_checks.csv	Different run and component numbering; same target.
R5	S2_fac_C0_s25_final_x4 and _x1 (audit v2)	Check G04: chain terminal on the planner's main component; P01: planner evidence for houses	Per scenario the audit lists the C0002 attachment as FAIL under G04 and P01:house (see the audit summary)	_connectivity_audit_v2/connectivity_summary.csv; audit_summary.md	The audit counts the C0002 attachment as FAIL. The earlier note '31 passed, 0 failed' does not match the file; the file is followed.
R6	S2_fac_C0_s25_final_x4 and _x1 (acceptance tests)	Per-run acceptance checks	0 FAIL; the off-main attachment is INFO	acceptance_tests.csv	The acceptance checks rate the off-main attachment as INFO, the audit as FAIL: two severity rules applied to the same data, not two different data.

No.	Run	Metric and definition	Observed	Evidence file	Reading
R7	Cause of component 10	Chain of road-line cells between component 10 and the main component	36 road cells in the minimum blocked chain: 33 outside the polygon and 3 inside within 8.4 m of it. Nearest main-component cell is 155.24 m away in the road-cell diagnostic and 151.55 m in the audit gate note	_boundary_mask_diag_v1 ; _chain_cell_reasons_v1 (verified baseline run)	Consistent with boundary truncation of the planning domain. The difference between the two distances is not reconciled. Whether the roads connect in reality is unknown.

The disconnection is explained in the road-cell and boundary diagnostics (run on the verified baseline) as a boundary-truncation effect: the shortest chain of road cells that would join component 10 to the main component has 36 cells, 33 outside the village polygon and 3 inside within 8.4 m of it, and cells outside the polygon are masked as impassable. This is an interpretation of planning-grid evidence. The roads outside the polygon were not checked for existence or usability, the polygon itself was not independently validated, and no route was planned outside it. The audit and acceptance validator classify the same route differently (FAIL and INFO), and the distance from component 10 to the main component differs between two diagnostics (155.24 m and 151.55 m); neither difference is resolved here. The issue is therefore open, not closed. On the TGRAC road lines, which also cover the part outside the polygon, the same road is connected to the main network.

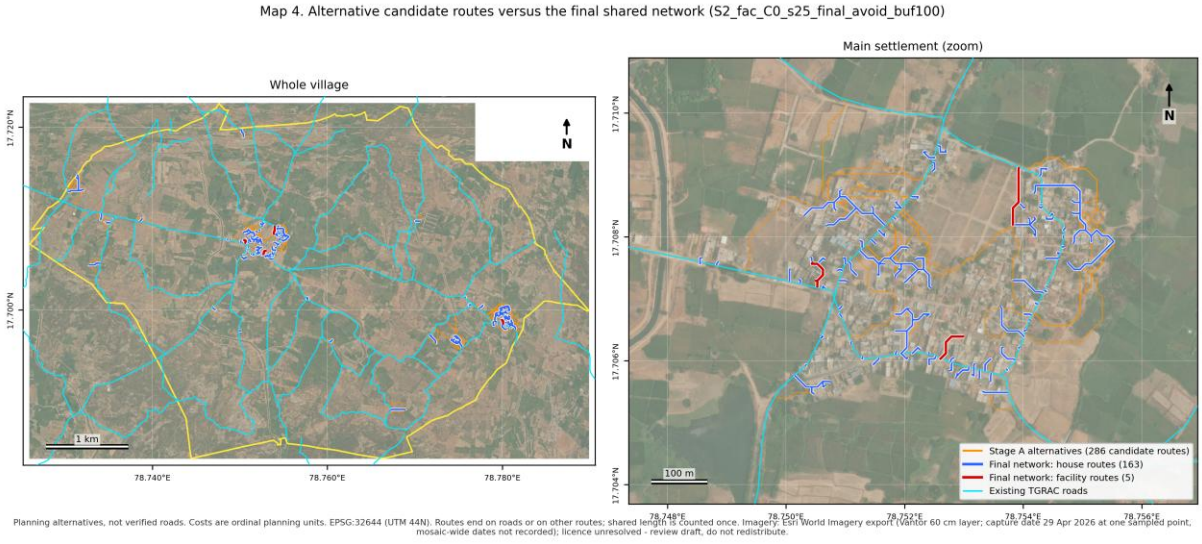


Figure 5. Damarakunta: alternative candidate routes versus the final shared network (map 4 of the final package; draft, model-derived; imagery licence unresolved).

9.6 Discussion

The results support C2 as a working, reproducible planner at planning level: shared-edge accounting lowers unique length and the building penalty trades some length for fewer footprint crossings. The C1 experiments are run on the real village and show the plan is fairly stable under synthetic omissions and additions, but they do not measure real segmentation

error. Adding administrative utilities (Gram Panchayat headquarters points) changes the house network slightly because houses can attach to the new facility routes, which shows that the facility set matters for the baseline numbers. The results do not support claims about cost in rupees, constructability or real-world connectivity, and C3 is untested.

9.7 Building-Avoiding Network and Illustrative Economics

Strict no-go zones around building footprints were tried first and failed: every house point lies inside its own blocked cell, so all 163 house routes failed. Footprints buffered by 7.1 m with a large penalty were then used. All 163 houses are routed, but the routes do not completely skirt buildings: the remaining crossings occur mostly where a target starts among touching buildings. Table 18 gives the effect (three retained facilities). The effect saturates by a penalty of x20.

Table 18. Building avoidance: length and footprint crossings (house routes; anganwadi excluded).

Scenario	Houses routed	House unique length (m)	Saving vs independent (%)	Routes crossing other footprints	Crossing length (m)	Facility routes (m)
x1, no penalty (P3 control)	163	4705.4	32.0	56	660.1	235.2
x4, penalty on footprint cells (primary)	163	5069.0	24.9	28	209.1	249.3
buffer 7.1 m, penalty x20	163	6238.1	18.5	19	147.9	302.4
buffer 7.1 m, penalty x100	163	6190.3	18.9	19	144.8	322.9
buffer 7.1 m, penalty x1000	163	6190.3	19.2	19	144.8	322.9

Illustrative economics (assumptions, not project data: Rs 75 lakh/km supplied by the project owner and not checked against PMGSY or a State Schedule of Rates; one assumed pavement section). The fair baseline is the unique length of the independent routes (6753.1 m), not their sum (10045.4 m), because a shared edge is built once. The saving is 1684.1 m (24.9%), about Rs 126.3 lakh at the assumed rate. Subtracting the shared network from the double-counted sum gives 4976.4 m (4.98 km), which overstates the saving about three times and is not a saving. These figures are not an engineering estimate.

Table 19. Illustrative length and cost saving of the shared network (Rs 75 lakh/km assumed).

Baseline	Baseline length (m)	Shared network (m)	Length saving (m)	Saving (%)	Cost saving (Rs lakh)
A (fair): independent routes, unique length	6753.1	5069.0	1684.1	24.9	126.3
B (double-counted): independent routes, total length	10045.4	5069.0	4976.4	49.5	373.2

10. LIMITATIONS AND THREATS TO VALIDITY

10.1 Evidence-Status Table

Table 20 lists each open issue, what is known, what is not, the corrective action and a closure criterion. An issue stays unresolved until the closure criterion is met; describing it in this report does not close it. The Damarakunta completion gate is not approved, and this report is not final.

Table 20. Evidence status of open issues.

Issue (type)	Known and evidence	Remaining unknown	Corrective action and closure criterion	Status
Imagery provenance, date, licence (data issue)	Esri World Imagery export of 4 Oct 2026, 11 requests, 0.6 m (DAMARAKUNTA_STAGE_A_INPUT_VALIDATION.md).	Capture date; licence terms for thesis figures and derived footprints.	Send the licence enquiry; record the answer and the capture date. Closed when a written answer or licence text is filed.	Unresolved
Facility identity and coordinates (data issue)	5 targets (3 school and health points, 2 Gram Panchayat HQ points); school layers attribute-only; Gram Panchayat building not identified; sub-centre has two TGRAC records 235.7 m apart; the anganwadi was excluded (facility_status.csv, inputs/EXCLUSIONS.md).	Which record is correct and where the entrance is.	Check against an official registry or imagery. Closed when each facility has a confirmed name and position.	Unresolved
Building-target uncertainty and merges (validation issue)	278 candidates; 84 unresolved merges; no reference labels (Stage A validation).	Detection accuracy; how many targets are real houses.	Label a village subset and score the detections. Closed when precision and recall are reported.	Unresolved
Missing land-cover and slope values (data issue)	178 land-cover and 961 slope gaps inside the polygon, a limitation of the 30 m source data, not interpolated; 29 of 32 interior blocked road cells fall on masked 30 m DEM cells.	Cause for 3 of those 32 cells; effect on routes.	Inspect DEM clip against the polygon (project item U08). Closed when every gap has a stated cause.	Partly explained
Disconnected road components and main-component reachability (data and domain limit)	12 components; target C0002 ends on a 5-cell component (Table 17).	Whether the road layer joins outside the grid.	Review imagery and the road layer beyond the boundary, with the guide's approval before changing the domain. Closed when the link is confirmed or the target is reported as unserved.	Unresolved
TGRAC road completeness (data issue)	44.70 percent of the raw road length lies outside the polygon (the earlier statement that all roads were inside is contradicted); the layer was clipped with gpd.clip() and only the 51,707.25 m inside the boundary is routed.	Whether the layer misses roads or includes wrong ones.	Compare with imagery for a sample. Closed when a completeness figure exists.	Unresolved
Parcel and school-compound exposure (fundamental limit)	Parcel layer covers 97.81 percent of the boundary; exposure counts are crossings of parcel units.	Ownership and legal access.	None within scope; keep as proxy.	Interpretation limit
Physical access, ownership, drainage, road width, money cost (outside scope)	No width or money parameter exists; costs are ordinal.	All of these.	Outside this deliverable; not estimated or invented.	Not in scope
Primary x4 versus P3 x1 reconciliation (validation issue)	Table 17 rows R1 to R3; unique house lengths 5,069.0 m (x4) and 4,705.4 m (x1) read from evaluation_summary.json of the final-family runs; the route-level P3 reproduction (U01) has not been run for this family.	Whether P3 reproduces from scratch.	Run the prepared reproduction. Closed when lengths match to 0.01 m.	Partly verified
Final high-resolution map exports (validation issue)	Four maps exist at 300 dpi; one C1 chart legend overlaps a label (U12).	Visual sign-off of each map.	Regenerate the chart and review each map. Closed when the guide accepts them.	Partly complete

Issue (type)	Known and evidence	Remaining unknown	Corrective action and closure criterion	Status
Literature survey completion (validation issue)	35 IEEE Transactions papers (2020 to 2026): metadata verified against the DOI registry; objective, method, findings and limitations extracted for every paper (Appendix A, Tables 2 to 4).	Novelty must still be argued against prior art beyond the 35 papers reviewed.	Extend the novelty comparison in Phase-II. Review accepted by the guide.	Complete
C3 transfer (blocked)	Not started.	All transfer results.	Written approval of the Damarakunta gate, then five villages. Closed by results.	Not started

10.2 Limitations

- No field validation, official confirmation or independent ground truth for buildings, facilities or roads.
- Facility identity is not verified against an official registry; facility weights are equal and provisional.
- The TGRAC road layer is generalised and incomplete, and licence terms are not recorded.
- The 30 m DEM resampled to a 10 m slope grid has no true 10 m terrain detail. The 961 slope gaps and 178 land-cover gaps inside the boundary are a spatial limitation of the 30 m source data and are not interpolated.
- Ordinal model costs are not monetary; successful execution is not feasibility.
- Cadastral geometry was repaired; parcel intersection is not ownership or access.
- C1 built-up cost 3.0 and the building-penalty set are provisional choices.
- Imagery date and licence, georeferencing accuracy and segmentation-model weights provenance are undocumented.
- Literature findings and limitations are project-oriented assessments of each paper's published methodology, not statements by the authors.
- Synthetic perturbations are sensitivity probes and do not measure real segmentation error.
- In nine of the ten add-perturbation runs some requested synthetic buildings could not be placed, so the perturbation size is slightly below the nominal fraction.

10.3 Methodological Limitations and Future Scope

10.3.1 Data sources, attribution and usage terms (status as of 2026-10-31)

The terms below are what the project has recorded. None has been confirmed in writing, and nothing here is legal advice or a statement that academic use, redistribution or publication of the data or of figures made from it is permitted.

Table 21. Data sources, attribution and recorded usage terms (none confirmed in writing).

Dataset	Source and attribution to cite	Recorded terms and status	Action before publication
TGRAC village boundary, 43 road lines, facilities (school and health layers; Gram Panchayat headquarters points from layer 10; the anganwadi record is excluded), cadastral parcels, school compounds	TGRAC, Government of Telangana (name as used in the project); ArcGIS REST services at tgrac.telangana.gov.in (boundary: Administrative Information System MapServer layer 27; roads: Schools MapServer layer 20); downloaded 4 to 5 Oct 2026	No licence or terms of use were found for these services as recorded. Currentness, completeness and authority are not verified. The saved boundary equals the stored query response (IoU 1.0), which shows consistency, not authority	Ask TGRAC in writing for terms for research use and for publishing derived maps; record the query date; cite as TGRAC, Government of Telangana
Esri World Imagery 0.6 m mosaic (11,296 x 7,171 px)	Esri World Imagery. The identify result saved with the project names Vantor (60 cm layer, dated 29 Apr 2026 at the one point sampled) and Earthstar Geographics as sources; attribution to Esri and its data providers is required by the terms summary the project recorded	The recorded Esri terms summary (21 Apr 2025) says basemap data may only be taken offline using Esri Content Packages, prohibits systematic harvesting of tiles by other methods, and requires attribution. The mosaic was built from 11 map-image export requests (4 Oct 2026), not Content Packages. Whether this use, model inference on the pixels, and publishing derived maps or figures is allowed is not determined. The enquiry draft has not been sent. Acquisition dates elsewhere in the mosaic are not recorded	Send the enquiry (or ask CBIT's GIS office whether an ArcGIS licence covers it). Until answered, do not redistribute the imagery or figures containing it; maps with imagery are review drafts
Building footprints (278 candidates)	Project extraction from the imagery above	Derived from imagery of unresolved terms, so the same question applies	Settle the imagery terms first
ESA WorldCover	ESA WorldCover (esa-worldcover.org); project notes record 2021 v200, not stored in the raster	CC BY 4.0 per project notes (not re-checked)	Confirm the product version and cite as the product requires
DEM (30 m, interpolated to 10 m)	Source not recorded; the file name n17_e078_1arc_v3 suggests an SRTM-type 1 arc-second product	Unknown	Identify the product and its terms
Google Open Buildings v3 (agreement checks only)	Sirko et al., 2021	CC BY 4.0 / ODbL per Google's notes (not re-checked)	Cite
LoveDA (segmentation benchmark only)	Wang et al., NeurIPS 2021 Datasets and Benchmarks; doi 10.5281/zenodo.5706578	CC BY 4.0 per its Zenodo record (checked 29 Oct 2026)	Cite

10.3.2 Field validation status

Every route in this work is the output of a GIS simulation: an ordinal cost surface on a 10 m raster, built from ESA WorldCover land cover and slope computed from a DEM of about 30 m resolution that was interpolated to 10 m, with the TGRAC road lines as the existing network and automatically detected building footprints as targets. The routes are unverified planning alternatives (status UNVERIFIED_PLANNING_ALTERNATIVE_NOT_A_ROAD).

No ground truthing has been done. There has been no field visit, no surveyed road centreline, width, drainage or soil data, no check of ownership or right of way, and no confirmation of facility positions or of building use. Interpolating a 30 m DEM does not give 10 m terrain accuracy. Ground-level physical validation is designated for future phases.

Proposed validation steps (not scheduled):

1. Verify facility positions against a registry, Google Earth and then the field; re-include the excluded anganwadi once its location is verified; locate the MeeSeva and Directorate of ESD (Aadhaar/service) centres and add them through inputs/admin_utilities_intake.csv; confirm which building is the Gram Panchayat office.
2. Label a subset of buildings in the village to measure detector precision and recall, and to replace the synthetic C1 omissions with measured errors.
3. Walk a sample of the proposed routes and record obstructions, width, gradient, drainage and crossings.
4. Check parcel and right-of-way status with revenue records.
5. Compare the DEM slope with a surveyed or higher-resolution profile.
6. Replace the illustrative cost and material assumptions with State Schedule of Rates values and a design section.

10.3.3 Network generalisation (C3)

C3 asks whether the method carries over to other villages. It has not started. Only Damarakunta has been processed, and no transfer result exists or is claimed. Testing across multiple villages remains part of the extended future work and begins only after the project owner approves the Damarakunta completion gate in writing. Planned, not scheduled: five additional villages, the same pipeline and metrics, a per-village input check (imagery, boundary, roads, facilities, DEM) and the licence questions of Section 13.1 settled first. Results from Damarakunta, for example the unique-length saving of the shared network, must not be quoted as general.

10.3.4 Other methodological limits

- Two Gram Panchayat headquarters points are included as administrative utilities; the MeeSeva and ESD centres are not, because no coordinate source exists.
- Costs are ordinal planning units, not rupees; the economic and material figures are assumption-driven and illustrative.

- The planner is a greedy heuristic with no optimality guarantee; shared-edge credit counts identical grid edges only.
- Building detections are unverified and the 163 routing targets are not a house count; agreement with Open Buildings is not accuracy.
- The omit/add runs (C1) are synthetic probes, not measured segmentation error.
- Building avoidance is bounded, not complete: some house routes still cross a neighbouring footprint (counts in Table 6), mostly where a target starts among touching buildings.
- 44.7% of the raw TGRAC road length lies outside the village polygon. The road layer was clipped to the boundary with gpd.clip() (51,707 m of 93,497 m remain, in 50 parts) and only the clipped roads are routed; the boundary itself has not been independently validated.
- The route of the health sub-centre uses a manual local override: the building penalty is lifted within 20 m of the facility point because the facility sits at the edge of its own footprint. The override changes no other route and is not a verified finding about access.
- 961 slope cells and 178 land-cover cells inside the boundary have no value because of the coarse 30 m DEM and land-cover source. This is a spatial limitation: the gaps are not filled by interpolation, and the planner treats them as invalid cells.

The sources, attribution and usage terms of every dataset are summarised in the Data Availability Statement before the References; no usage terms have been confirmed in writing.

11. FUTURE WORK AND PHASE-II PLAN

1. Replace the synthetic perturbations by label-based error propagation: build reference labels for a village subset and measure how real segmentation errors change the plan (C1).
2. Check the road fragment on which target C0002 ends against imagery and the road layer beyond the boundary, and review the maps and inset boxes visually.
3. Verify facility identities against official registries and obtain imagery metadata.
4. Document the DEM and land-cover sources and test a finer DEM where available.
5. Decide whether a second village allows an honest test of C3.
6. Extend the novelty comparison with prior art beyond the 35 IEEE Transactions papers reviewed.
7. Run the P3 baseline reproduction and send the imagery licence enquiry.

12. CONCLUSION

This Phase-I report framed a planning-level pipeline for rural access networks in Damarakunta, completed a literature review of 35 IEEE Transactions papers with five foundational references, documented the datasets and their limits, and presented preliminary results from a two-stage planner with shared-edge credit. The planner routes all 163 eligible targets at the 25° slope limit and reduces unique network length compared with independent routing. The outputs are model-derived and unvalidated, the C1 experiments are synthetic sensitivity probes without real labels, C3 transfer remains to be done, and the Phase-II plan targets both together with verification of data sources.

DATA AVAILABILITY STATEMENT

The study uses publicly accessible geospatial data for Damarakunta village, Siddipet district, Telangana, for non-commercial academic research only.

Satellite imagery. Imagery was obtained through the Esri World Imagery service, accessed via its public ArcGIS REST export endpoint (services.arcgisonline.com). The imagery is © Esri and its data providers and is used here under Esri's published terms for the service. The imagery itself is not redistributed with this report.

Administrative and infrastructure data. The village boundary, road network, cadastral parcels, school and health facility locations and Gram Panchayat headquarters points were obtained from the Telangana Remote Sensing Applications Centre (TGRAC), Government of Telangana, through its public ArcGIS REST services (tgrac.telangana.gov.in). Ownership of these data remains with the originating agency.

Other data. Elevation and land-cover rasters were taken from publicly available global products, as documented in the data-sources table (Section 10.3).

Derived data and code. Building footprints, cost surfaces, routes and result tables were generated by the author from these inputs and are model-derived, planning-level outputs. The processing scripts, run manifests with file hashes and result tables are available from the author on reasonable request. Redistribution of the source layers is subject to the terms of the providers named above.

Written confirmation of usage terms will be requested from the data providers; until it is received, no redistribution rights are claimed.

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A. VERIFIED BIBLIOGRAPHIC DETAILS OF THE 35 IEEE TRANSACTIONS PAPERS

Each entry below was checked against its DOI-registry record (Crossref) on 29 October 2026: all authors, journal, volume, issue where given, pages or article number, year and DOI. Every journal is an IEEE Transactions title and every paper was published between 2020 and 2026. Links of the form https://doi.org/DOI lead to the IEEE Xplore page. Together with the objective, method, findings and limitations extracted for each paper (Tables 2 to 4), this completes the literature review for the 35-paper scope.

Table 22. Verified details of the 35 qualifying IEEE Transactions papers.

Ref.	Authors	Journal, volume, pages, year	DOI
[1]	Qingyu Li, Lichao Mou, Yao Sun, Yuansheng Hua, Yilei Shi, Xiao Xiang Zhu	IEEE Trans. Geoscience and Remote Sensing, vol. 62, pp. 1-15, 2024	10.1109/TGRS.2024.3369723
[2]	Ailong Ma, Chenyu Zheng, Junjue Wang, Yanfei Zhong	IEEE Trans. Geoscience and Remote Sensing, vol. 61, pp. 1-17, 2023	10.1109/TGRS.2023.3265186
[8]	Yinghui Xing, Dexuan Kong, Shizhou Zhang, Ziyi Li, Qingyi Li, Yanning Zhang	IEEE Trans. Geoscience and Remote Sensing, vol. 64, pp. 1-16, 2026	10.1109/TGRS.2025.3647015
[9]	Junjie Fu, Chenliang Wang, Mingzhe Liu, Xinghua Li, Yi Liu, Wenjiao Shi, Ruili Wang	IEEE Trans. Geoscience and Remote Sensing, vol. 64, art. no. 5620016, 2026	10.1109/TGRS.2026.3687072
[10]	Xu Wang, Luyao Wang, Wenxin Huang, Yibo Liu, Sheng Zhang, Xian Zhong	IEEE Trans. Geoscience and Remote Sensing, vol. 63, pp. 1-10, 2025	10.1109/TGRS.2025.3598813
[11]	Peng Liu, Xin Gao, Chaojun Shi, Yiguo Lu, Lu Bai, Yingying Fan, Yalong Xing, Yurong Qian	IEEE Trans. Geoscience and Remote Sensing, vol. 63, pp. 1-12, 2025	10.1109/TGRS.2025.3593017
[12]	Shanwen Wang, Xin Sun, Changrui Chen, Danfeng Hong, Jungong Han	IEEE Trans. Geoscience and Remote Sensing, vol. 63, pp. 1-15, 2025	10.1109/TGRS.2025.3585489
[13]	Qinglong Cao, Yuntian Chen, Chao Ma, Xiaokang Yang	IEEE Trans. Geoscience and Remote Sensing, vol. 63, pp. 1-14, 2025	10.1109/TGRS.2025.3559557
[14]	Lilu Zhu, Kai Huang, Yang Wang, Yanfeng Hu	IEEE Trans. Intelligent Transportation Systems, vol. 26, no. 11, pp. 19163-19181, 2025	10.1109/TITS.2025.3589586
[15]	Yuchuan Wang, Ling Tong, Shiyu Luo, Fanghong Xiao, Jiaxing Yang	IEEE Trans. Geoscience and Remote Sensing, vol. 62, pp. 1-18, 2024	10.1109/TGRS.2024.3379988
[16]	Wenna Wang, Lingyan Ran, Hanlin Yin, Mingjun Sun, Xiuwei Zhang, Yanning Zhang	IEEE Trans. Geoscience and Remote Sensing, vol. 62, pp. 1-13, 2024	10.1109/TGRS.2024.3373493
[17]	Haonan Guo, Xin Su, Chen Wu, Bo Du, Liangpei Zhang	IEEE Trans. Geoscience and Remote Sensing, vol. 62, pp. 1-16, 2024	10.1109/TGRS.2024.3383057
[18]	Ruoyu Yang, Yanfei Zhong, Yinhe Liu, Xiaoyan Lu, Liangpei Zhang	IEEE Trans. Geoscience and Remote Sensing, vol. 62, pp. 1-16, 2024	10.1109/TGRS.2024.3387945
[19]	Yuxuan Wang, Lanxin Zeng, Wen Yang, Jian Kang, Huai Yu, Mihai Datcu, Gui-Song Xia	IEEE Trans. Geoscience and Remote Sensing, vol. 62, pp. 1-15, 2024	10.1109/TGRS.2024.3359704
[20]	Qian Tang, Yapeng Li, Yongchao Xu, Bo Du	IEEE Trans. Geoscience and Remote Sensing, vol. 62, pp. 1-14, 2024	10.1109/TGRS.2024.3505595
[21]	Daoyuan Zheng, Shaohua Wang, Haixia Feng, Shunli Wang, Mingyao Ai, Pengcheng Zhao, Jiayuan Li, Qingwu Hu	IEEE Trans. Geoscience and Remote Sensing, vol. 62, pp. 1-15, 2024	10.1109/TGRS.2024.3438248
[22]	Zhibao Wang, Huan Chang, Lu Bai, Liangfu Chen, Xiuli Bi	IEEE Trans. Geoscience and Remote Sensing, vol. 62, pp. 1-13, 2024	10.1109/TGRS.2024.3477749
[23]	Jinpeng Li, Jun He, Weijia Li, Jiabin Chen, Jinhua Yu	IEEE Trans. Geoscience and Remote Sensing, vol. 62, pp. 1-18, 2024	10.1109/TGRS.2024.3380914
[24]	Zhi Zheng, Shunyu Yao, Genghui Li, Linxi Han, Zhenkun Wang	IEEE Trans. Intelligent Transportation Systems, vol. 25, no. 1, pp. 1033-1043, 2024	10.1109/TITS.2023.3313688
[25]	Zhen Wang, Xinwei Shen, Hongbin Sun, Qiuwei Wu	IEEE Trans. Power Systems, vol. 39, no. 6, pp. 7038-7049, 2024	10.1109/TPWRS.2024.3384534
[26]	Reo Iizuka, Junshi Xia, Naoto Yokoya	IEEE Trans. Geoscience and Remote Sensing, vol. 62, pp. 1-14, 2024	10.1109/TGRS.2023.3344670
[27]	Chenbin Liang, Weibin Li, Yunyun Dong, Wenlin Fu	IEEE Trans. Geoscience and Remote Sensing, vol. 62, pp. 1-16, 2024	10.1109/TGRS.2024.3379669
[28]	Jingtao Hu, Junyu Gao, Yuan Yuan, Jocelyn Chanussot, Qi Wang	IEEE Trans. Geoscience and Remote Sensing, vol. 61, pp. 1-12, 2023	10.1109/TGRS.2023.3305031
[29]	Xiaoqian Zhu, Xiangrong Zhang, Tianyang Zhang, Xu Tang, Puhua Chen, Huiyu Zhou, Licheng Jiao	IEEE Trans. Geoscience and Remote Sensing, vol. 61, pp. 1-13, 2023	10.1109/TGRS.2023.3317080
[30]	Jinming Zhang, Xiangyun Hu, Yujun Wei, Lili Zhang	IEEE Trans. Geoscience and Remote Sensing, vol. 61, pp. 1-13, 2023	10.1109/TGRS.2023.3241679

Ref.	Authors	Journal, volume, pages, year	DOI
[31]	Qianxiong Xu, Cheng Long, Liang Yu, Chen Zhang	IEEE Trans. Geoscience and Remote Sensing, vol. 61, pp. 1-14, 2023	10.1109/TGRS.2023.3261332
[32]	Lei Ding, Hao Tang, Yahui Liu, Yilei Shi, Xiao Xiang Zhu, Lorenzo Bruzzone	IEEE Trans. Image Processing, vol. 31, pp. 678-690, 2022	10.1109/TIP.2021.3134455
[33]	Qingyu Li, Lichao Mou, Yuansheng Hua, Yilei Shi, Xiao Xiang Zhu	IEEE Trans. Geoscience and Remote Sensing, vol. 60, pp. 1-17, 2022	10.1109/TGRS.2021.3109844
[34]	Yao Sun, Yuansheng Hua, Lichao Mou, Xiao Xiang Zhu	IEEE Trans. Geoscience and Remote Sensing, vol. 60, pp. 1-15, 2022	10.1109/TGRS.2020.3043089
[35]	Yuan Zhou, Zhanlong Chen, Bin Wang, Shuangjiang Li, Hao Liu, Daozhu Xu, Chao Ma	IEEE Trans. Geoscience and Remote Sensing, vol. 60, pp. 1-17, 2022	10.1109/TGRS.2022.3152575
[36]	Yu Liu, Zhengyang Zhao, Shanwen Zhang, Lei Huang	IEEE Trans. Geoscience and Remote Sensing, vol. 60, pp. 1-10, 2022	10.1109/TGRS.2022.3192614
[37]	Hoin Jung, Han-Soo Choi, Myungjoo Kang	IEEE Trans. Geoscience and Remote Sensing, vol. 60, pp. 1-12, 2022	10.1109/TGRS.2021.3108781
[38]	Bingnan Li, Jiuchong Gao, Shuiping Chen, Samsung Lim, Hai Jiang	IEEE Trans. Geoscience and Remote Sensing, vol. 60, pp. 1-16, 2022	10.1109/TGRS.2022.3174399
[39]	Bo Zhang, Tao Chen, Bin Wang	IEEE Trans. Geoscience and Remote Sensing, vol. 60, pp. 1-12, 2022	10.1109/TGRS.2021.3117851
[40]	Jie Mei, Rou-Jing Li, Wang Gao, Ming-Ming Cheng	IEEE Trans. Image Processing, vol. 30, pp. 8540-8552, 2021	10.1109/TIP.2021.3117076